Markov-Switching Multivariate GARCH Model with Copula-Distributed Innovations

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Abstract

This paper introduces a novel Markov-switching multivariate GARCH model with copula innovations, designed to enhance the dynamic assessment of financial market interdependencies. The model captures potential regime shifts in asymmetric tail dependence structures among volatility innovations in speculative return series. We propose a computationally efficient two-step quasi-maximum likelihood estimator that is both consistent and asymptotically normal under standard conditions. Monte Carlo simulations further demonstrate its favorable finite-sample properties. An empirical application to a bivariate system of log-returns of NASDAQ Composite and Bitcoin shows that the Markov-switching model effectively uncovers shifts in tail dependence patterns, which remain undetected or averaged out in Gaussian or time-invariant copula-MGARCH models. The best-fitting three-state specification reveals a shift to a lower-tail dependence state both in the lead-up to and during the COVID-19 pandemic, indicating that the returns of both assets become increasingly intertwined during market downturns, exposing them to large, simultaneous negative shocks and amplifying the risk of severe joint declines. The findings suggest that, rather than providing diversification benefits, Bitcoin may exacerbate losses during periods of crisis.

Keywords: Copula, multivariate GARCH, Markov-switching, hedging, Bitcoin, NASDAQ

JEL Classification: C32, C58.

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1 Introduction

Modeling the volatility dynamics of speculative return processes represents a fundamental problem in financial econometrics. Over the years, univariate generalized autoregressive conditional heteroskedasticity (GARCH) (Bollerslev, 1986) and multivariate GARCH (MGARCH) models have emerged as standard tools in both academic research and practical applications for capturing key features of financial time series, such as volatility clustering and spillover effects (for an overview see, for example, Bauwens et al., 2006). Among the various MGARCH specifications, the BEKK model (Engle and Kroner, 1995) and the DCC model (Engle, 2002) have become particularly popular for dynamic covariance modeling in small- to medium-sized systems. Despite their success, standard MGARCH setups have been criticized for their lack of capturing the complex higher-order dependencies observed in the covariance innovations, which are typically assumed to be i.i.d. Gaussian. Empirical studies by Erb et al. (1994); Longin and Solnik (2001); Ang and Chen (2002) and Hsu et al. (2008) highlight the violation of independence assumptions, while Patton (2004) points to the presence of tail dependencies in financial returns during periods of crisis. In response to these limitations, researchers have turned to more flexible modeling approaches that go beyond the restrictive Gaussian assumption. Copula functions, introduced by Sklar (1959), offer a versatile framework for modeling a wide range of dependence structures between random variables. Notably, Lee and Long (2009) developed the copula-MGARCH (C-MGARCH) framework, which allows for the simultaneous yet separate modeling of conditional correlations and dependence structures, and thus preserves the ability to capture volatility spillovers across different assets. The empirical advantages of C-MGARCH, particularly in terms of information criteria and value-at-risk (VaR) forecasting, have been further demonstrated by Christoffersen et al. (2012); Fengler et al. (2012) and Herwartz and Raters (2015).[1] Several recent studies have proposed combining marginal stochastic volatility models with single-factor copulas (Kreuzer and Czado, 2021) or employing inversion copulas (Loaiza-Maya and Smith, 2020). However, these models generally fall short in accounting for volatility spillovers and fail to incorporate time-varying dependence structures, as they rely on univariate stochastic volatility specifications and static copulas, respectively.

Beyond addressing higher-order dependencies through copula distributions, another important extension to MGARCH models lies in accounting for time-varying patterns in the dependence structure, in particular using Markov switching techniques (Hamilton, 1989, 2016). Classical hidden Markov models (HMMs) primarily focus on shifts in volatility regimes, and their utility has been demonstrated in various applications, including risk assessment and improved model fit in portfolio management (Gray, 1996; Klaassen, 2002; Bauwens

[1] Unlike the C-MGARCH model, other approaches to include copulas into GARCH frameworks - for instance, Hsu et al. (2008) - do not capture volatility spillovers. For an overview of different stochastic volatility models including copulas, we refer the reader to Patton (2012).

et al., 2010; Ardia et al., 2018). However, existing Markov switching (M)GARCH models do not capture shifts in the dependence structure between different financial assets. Introducing Markov switching into the copula component of C-MGARCH models offers the potential to improve both the goodness-of-fit and risk forecasting by allowing for dynamic (asymmetric) dependence patterns that evolve over time (Ang and Chen, 2002). While some efforts have been made to combine univariate GARCH models with regime-switching copulas, they remain limited in scope. One approach, advocated by Jondeau and Rockinger (2006), Garcia and Tsafack (2011), and Wang et al. (2011), incorporates state-dependent copulas to model alternating dependence structures, while another approach, suggested by Chollete et al. (2009), models copula parameters as following a Markov chain. Additionally, Liu et al. (2017) introduced a framework that integrates various copulas and time-varying copula parameters, although their model does not employ a Markov chain for time dependence. These studies have uncovered important evidence of time-varying, asymmetric dependencies between financial assets. However, they primarily rely on univariate GARCH (or stochastic volatility) models, which fail to account for volatility spillovers between asset classes, such as equities, commodities, and bonds. Similarly, MGARCH jump models proposed by Drost and Werker (1996) address changing dependency patterns, specifically switching higher-order moments, but they suffer from computational complexity due to their larger parameter space, and they incorporate tail dependencies only indirectly.

This paper presents a novel framework that incorporates copula-distributed covariance innovations into MGARCH models, with the copula state space governed by a discrete Markov chain. We refer to this model as the Markov-switching copula-MGARCH model (MS-C-MGARCH). Building on the C-MGARCH model of Lee and Long (2009), we extend their approach by allowing for state-dependent joint error distributions, thus accommodating time-varying strengths and degrees of dependencies, as well as potential breaks in the marginal distributions.[2] To strike a balance between model flexibility and parsimony, we maintain a fixed law of motion for the conditional covariance matrix across regimes. This specification avoids issues of path dependence while allowing us to derive a computationally tractable two-step quasi maximum likelihood (ML) estimation procedure, which is both consistent and asymptotically normal under standard assumptions for BEKK and HMM models (see, e.g. Hafner and Preminger, 2009; Leroux, 1992). The estimation method is straightforward to implement and can be easily incorporated into existing software packages for MGARCH model estimation such as Fülle et al. (2022). Our Monte-Carlo simulation results confirm the consistency of the proposed two-step estimator in large samples and demonstrate its robust performance even for moderately sized time series, such as those containing 1,000 or 2,500 observations. Unsurprisingly, the finite sample performance of the estimator depends on the accuracy of state assignment and the strength of the

[2]Lee and Long (2009) introduced time-varying copula parameters, but their framework does not account for shifts between different copula structures beyond the given copula and its parameterization.

underlying higher-order dependence patterns. In particular, transitory states prove more challenging to detect, while estimation accuracy improves significantly when state-specific higher-order dependencies exhibit marked heterogeneity. Notably, the second step of the estimation process, which conditions on the first step's covariance estimates, experiences only a minimal efficiency loss. Hence, the proposed two-step estimator, despite the often large parameter space of MGARCH models, does not compromise estimation precision for computational feasibility when compared to one-step joint ML estimation of all model parameters. While our framework is flexible and applicable to various MGARCH specifications, we focus on the BEKK model due to its widespread use in the literature, as seen in Lee and Long (2009), and its advantages in terms of generality, positive-definiteness, and identification under mild conditions (Engle and Kroner, 1995).

To empirically illustrate the implementation and advantages of the MS-C-MGARCH model, we analyze a bivariate system comprising returns from the NASDAQ Composite index and Bitcoin. Model selection exercises reveal substantial improvements in goodness-of-fit with the three-state MS-C-MGARCH model, compared to two-state models and time-invariant benchmarks, including Gaussian and various copula-based MGARCH specifications. The best-performing three-state model uncovers compelling evidence of heterogeneous higher-order dependence patterns (i.e., independence, upper-tail dependence, and lower-tail dependence) and their dynamic evolution across different states, particularly in the wake of the global pandemic in early 2020. The superior performance of the MS-C-MGARCH model is intuitive, as the Gaussian benchmark fails to capture tail dependencies, while the time-invariant C-MGARCH model 'averages out' periods of distinct lower- and upper-tail dependencies. Our findings challenge the notion that Bitcoin serves as a hedge or offers diversification benefits against NASDAQ exposure. Instead, the estimation results point to volatility innovations shifting toward lower-tail dependence during market turmoil, indicating that holding Bitcoin can exacerbate losses during extreme market downturns. A striking example is the joint plummet of both assets on 'Black Thursday,' March 12, 2020, when the sharp positive correlation between returns, coupled with an almost unit probability of a state of lower-tail dependence, led to large, simultaneous negative shocks in both markets.

The remainder of this work is organized as follows. The next section introduces the new MS-C-MGARCH model, while Section 3 provides a comprehensive simulation study to investigate the finite-sample performance of the estimator proposed. Section 4 presents an empirical illustration that argues for viable changes of higher-order dependence patterns in a bivariate system of NASDAQ and Bitcoin returns. Section 5 concludes. The appendix collects the proofs. An online appendix (OA) includes a brief definition of copulas employed in this work, with a generalization towards higher dimensions (OA A), a detailed algorithm for the suggested two-step estimation procedure (OA B), detailed simulation results (OA C), and a graphical display of the analyzed return series along with descriptive statistics (OA D). Throughout the paper, $\|x\|$ is used to denote

the Euclidean norm for a finite-dimensional vector x, whereas for a bounded linear operator A, $\|A\| = \sup_{\|x\|\leq 1} \|Ax\|$ is the operator norm. Note that $\|A\|^2 = \rho(A'A) \leq \mathrm{tr}\,(A'A)$ where $\rho(.)$ denotes the spectral radius. For an N-dimensional square matrix A, $|A|$ denotes the determinant and $|a|$ is the absolute value if a is a scalar. Note that $|A| \leq \|A\|^N$. We use $B(x,\delta)$ to indicate an open ball centered at x with radius $\delta > 0$, and $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$ denote the non-negative and non-positive part of f, respectively.

2 Markov switching copula-MGARCH model

In this section, we first sketch the C-MGARCH model of Lee and Long (2009) and then generalize this model towards a regime switching specification. Let $\{r_t\}_{t\in\mathbb{N}}$, $r_t \in \mathbb{R}^N$, be a sequence of stochastic returns of N speculative assets and Ω_t be a natural filtration generated as $\Omega_t = \boldsymbol{\sigma}(\{r_u : 0 < u \leq t\})$. The multivariate GARCH model is

$$r_t = \mu_t + e_t, \tag{1}$$

$$e_t = H_t^{1/2}\eta_t, \tag{2}$$

where η_t is a strictly stationary martingale difference sequence such that $\mathbb{E}[\eta_t|\Omega_t] = 0$ and $\mathbb{E}[\eta_t\eta_t'|\Omega_t] = I_N$. Notably, μ_t and H_t are the conditional mean and covariance matrix of r_t, respectively, i.e., $\mathbb{E}[r_t|\Omega_t] = \mu_t$ and $\mathrm{Cov}[r_t|\Omega_t] = H_t$ with $H_t = H_t^{1/2}H_t^{1/2'}$. Without loss of generality, it is often assumed that $\mu_t = \mu$ and the empirical analysis is based on centered return processes such that $\mu = 0$. There exist different methods to specify the conditional covariance matrices, H_t. In particular, the BEKK representation proposed by Engle and Kroner (1995) is sufficiently general to accommodate most relevant joint second-order dynamic structures and has the benefit of guaranteeing the positive definiteness of the conditional covariance process under mild conditions. In this paper, we employ the the BEKK specification for H_t. Specifically, a BEKK(1,1,1) model reads as

$$H_t = QQ' + A'e_{t-1}e_{t-1}'A + G'H_{t-1}G, \tag{3}$$

with Q, A and G being $(N \times N)$-dimensional coefficient matrices, where Q is lower triangular. Accordingly, the vector of BEKK(1,1,1) parameters is $\phi = (\mathrm{vech}(Q)', \mathrm{vec}(A)', \mathrm{vec}(G)')' \in \Phi \subset \mathbb{R}^{\frac{N(N+1)}{2}+2N^2}$. Conditions for the existence of a strictly stationary solution and identification will be stated later. The unconditional distribution of the innovation η_t is often specified to be elliptical such as an N-dimensional multivariate Gaussian distribution. This rules out the possibility that components in η_t may exhibit tail dependence of

different forms, since a diagonal covariance implies independence for elliptical distributions. To address this issue, Lee and Long (2009) introduced the copula-MGARCH (C-MGARCH) model that permits dependence among the components of η_t while preserving the uncorrelatedness. We briefly describe the C-MGARCH next.

2.1 The copula-MGARCH model

In the C-MGARCH suggested by Lee and Long (2009), the law of η_t is specified as a standardized N-dimensional copula distribution. For notational convenience, we consider the stylized case of bivariate systems ($N = 2$) and discuss extensions towards higher dimensions in Section 2.5 and OA A.2 (see Joe, 1996; Nelsen, 2006, for textbook treatments). Let $\xi_{t,1}$ and $\xi_{t,2}$ be random variables with marginal CDF given by $F_i(\xi_{t,i};\theta_i)$, $i = 1, 2$. The joint CDF of random vector ξ_t is specified as

$$\xi_t \sim F(\xi_t;\theta) = C(F_1(\xi_{t,1};\theta_1), F_2(\xi_{t,2};\theta_2);\theta_c), \tag{4}$$

where C is a copula function with parameter θ_c and $\theta = (\theta_1', \theta_2', \theta_c)'$. Denote the conditional covariance matrix $\mathrm{Cov}\,[\xi_t|\Omega_{t-1}] = \Sigma$, which can be obtained from their marginals based on the Hoeffding's Lemma. Note that the matrix Σ also depends on θ. The innovation process in a C-MGARCH model is given by $\eta_t = \Sigma^{-1/2}\xi_t$. It is easy to verify that $\mathbb{E}[\eta_t\eta_t'|\Omega_{t-1}] = I_2$ and thus $\mathbb{E}[e_te_t'|\Omega_{t-1}] = H_t$. Suppose each element in ξ_t has a density function $f_i(x;\theta_i)$ with respect to the Lebesgue measure ν on $\mathbb{R}$, the joint PDF of ξ_t is given by

$$f(\xi_t;\theta) = f_1(\xi_{t,1};\theta_1)f_2(\xi_{t,2};\theta_2)c(F_1(\xi_{t,1};\theta_1), F_2(\xi_{t,2};\theta_2);\theta_c), \tag{5}$$

where $c(u_1, u_2;\theta) = \frac{\partial^2 C(u_1,u_2;\theta)}{\partial u_1 \partial u_2}$ is the copula density function. By a change of variable formula, the joint PDF of the GARCH innovation is $f_{\eta_t}(\eta_t;\theta) = |\Sigma^{1/2}|f(\Sigma^{1/2}\eta_t;\theta)$.

It is worth remarking that the law of the i-th element in η_t is a convolution of F_1 and F_2 and generally differs from F_i if C is not a product copula (i.e., $C(u_1, u_2;\theta) \neq u_1u_2$). Moreover, even if F_i are Gaussian, the distribution of η_t and hence, the return process, is in general non-Gaussian. More precisely, the MGARCH specification not only determines the conditional second moments of the C-MGARCH model but also the dependence structure. Multiplying η_t by $H_t^{1/2}$ leads to a time-varying combination of the elements of η_t and thus time-varying conditional marginals of the return process. In comparison with Gaussian MGARCH models, the C-MGARCH model provides enhanced flexibility in terms of conditional time-varying marginals going beyond first- and second-order moments. Nevertheless, in a static copula setting, the higher order

dependence structure among different assets in the return process is assumed to be time invariant. This study generalizes the C-MGARCH approach towards a Markov switching C-MGARCH model. We outline our approach next.

2.2 The Markov switching C-MGARCH model

Let $\{s_t\}_{t\in\mathbb{N}}$ be a Markov chain on finite state space $\{1,\dots,K\}$, known as *regimes*. For each $t \in \mathbb{N}$, the conditional distribution of ξ_t depends on the corresponding state s_t,

$$\xi_t\,(k)\,|s_t = k \sim F_k(\xi_t;\theta_k) = C_k(F_{k,1}(\xi_{t,1};\theta_{k,1}), F_{k,2}(\xi_{t,2};\theta_{k,2});\theta_{k,c}),\ k = 1,\dots,K. \tag{6}$$

Notably, the joint distribution is time-varying in the sense that both the marginals and the copula function, as well the parameters characterizing them are allowed to be state-dependent. For notational convenience, we write $\xi_t(k)$, $\Sigma(k)$, $\eta_t(k) = \Sigma(k)^{-1/2}\xi_t(k)$ and $e_t(k) = H_t^{1/2}\eta_t(k)$, if the state variable takes value $s_t = k$. Suppose $\xi_t(k)$ has a density with respect to ν^2 in all states, then similar to (5),

$$f_k(\xi_t;\theta_k) = f_{k,1}(\xi_{t,1};\theta_{k,1}) f_{k,2}(\xi_{t,2};\theta_{k,2}) c_k(F_{k,1}(\xi_{t,1};\theta_{k,1}), F_{k,2}(\xi_{t,2};\theta_{k,2});\theta_{k,c}),$$

where $\theta_k = \left(\{\theta'_{k,i}\}_{i=1}^N, \theta_{k,c}\right)' \in \Theta_k$ and $f_{k,i}$ and c_k are defined analogously. The joint PDF of $\eta_t(k)$ can be obtained accordingly as $f_{\eta_t(k)}(\eta_t;\theta_k) = |\Sigma(k)^{1/2}| f_k(\Sigma(k)^{1/2}\eta_t;\theta_k)$.

Remark 1. *The MS model outlined in (6) is flexible with respect to the copula (shape of dependence) function C_k, (strength of dependence) parameter $\theta_{k,c}$, the marginal distributions $F_{k,i}$ and their parameters $\theta_{k,i}$. In principle, these directions of state-dependence might be considered simultaneously or separately to form a MS-C-MGARCH model. In this work, we concentrate on state-specific copula functions C_k with parameters $\theta_{k,c}$ to account for changes in the shape and strength of dependence that one might expect for the relation between different asset classes (e.g. equity and safe-haven) under heterogeneous market conditions (normal times, crisis and recovery).*

Suppose the $\{s_t\}$ has a time-homogenous transition probability matrix, denoted by P such that

$$P = \begin{pmatrix} p_{11} & \cdots & p_{1K} \\ \vdots & \ddots & \vdots \\ p_{K1} & \cdots & p_{KK} \end{pmatrix}, \tag{7}$$

where $p_{ij} = \mathbb{P}(s_t = i | s_{t-1} = j)$ for $i, j \in \{1,\dots,K\}$, which are included in the parameter vector and will be

estimated from the observable data $\{r_t\}$. However, since transition probabilities in the matrix P sum up to one for each row, there are only $K(K-1)$ free parameters. Let ψ be the parameter vector that contains the shape parameters for both the marginals and copula functions in all states as well as transition probabilities,

$$\psi = \left(\{\theta_k'\}_{k=1}^K, \{\{p_{ij}\}_{i=1}^K\}_{j=1}^{K-1}\right)',$$

while the parameter vector to be estimated is $\kappa = (\phi', \psi')' \in \mathcal{K} = \Phi \times \Psi$ with $\Psi \subset \left(\times_{k=1}^K \Theta_k\right) \times (0,1)^K$ being the parameter space of ψ. As an illustrative sample, suppose the marginals $F_{k,i}$ are standard Gaussian for all k and i as in Lee and Long (2009), $\theta_{k,i}$ are known and $\theta_k = \theta_{k,c}$ reduces to a scalar. The parameter space Θ_k for each k is a compact subset of $(\underline{\theta_{k,c}}, \infty)$ with $\underline{\theta_{k,c}}$ being the infimum of the parameter support of the underlying copula function in state k.

Since the state space is finite, there exists a stationary distribution depending on ψ, denoted $\pi(k)$ for $k = 1, \ldots, K$, from which the initial state s_1 is drawn. Suppose that the initial conditional covariance H_1 is also generated from its stationary distribution, which exists as we show later under specific conditions, the joint density with respect to ν^T of the return series $(r_1, \ldots, r_T)'$ is

$$\begin{aligned} f_r\left((r_1, \ldots, r_T)'; \kappa\right) &= \sum_{s_1=1}^K \cdots \sum_{s_T=1}^K \pi(s_1) |\Sigma(s_1)^{1/2} H_1^{-1/2}| f_{s_1}\left(\Sigma(s_1)^{1/2} H_1^{-1/2} r_1\right) \\ &\quad \prod_{t=2}^T p_{s_t s_{t-1}} |\Sigma(s_t)^{1/2} H_t^{-1/2}| f_{s_t}\left(\Sigma(s_t)^{1/2} H_t^{-1/2} r_t\right) \\ &= \sum_{s_1=1}^K \cdots \sum_{s_T=1}^K \pi(s_1) |H_1^{-1/2}| f_{\eta_t(s_1)}\left(H_1^{-1/2} r_1\right) \prod_{t=2}^T p_{s_t s_{t-1}} |H_t^{-1/2}| f_{\eta_t(s_t)}\left(H_t^{-1/2} r_t\right), \end{aligned}$$

where the dependence of matrices H_t, Σ and density functions on parameters on the right-hand side is dropped to simplify the notation. The hidden state variable is inferred through a filter (Hamilton, 1989):

$$\mathbb{P}(s_t = i|\Omega_t) = \frac{\mathbb{P}(s_t = i|\Omega_{t-1}) f_{r_t(i)}(r_t|\Omega_{t-1})}{\sum_{k=1}^K \mathbb{P}(s_t = k|\Omega_{t-1}) f_{r_t(k)}(r_t|\Omega_{t-1})}, \tag{8}$$

where $\mathbb{P}(s_t = i|\Omega_{t-1}) = \sum_{j=1}^K p_{ij} \mathbb{P}(s_{t-1} = j|\Omega_{t-1})$ and $f_{r_t(i)}(r_t|\Omega_{t-1}) = |H_t^{-1/2}| f_{\eta_t(i)}\left(H_t^{-1/2} r_t\right)$. Note that conditional on the return series, $\{s_t\}$ is still a Markov chain, albeit no longer time-homogeneous. Using the

conditional probabilities of state variables, the joint density can be written as

$$
\begin{aligned}
f_r\left((r_1,\ldots,r_T)';\kappa\right) &= \prod_{t=1}^{T}|H_t^{-1/2}|\left[\sum_{k=1}^{K}|\Sigma(k)^{1/2}|f_k\left(\Sigma(k)^{1/2}H_t^{-1/2}r_t\right)\mathbb{P}(s_t=k|\Omega_{t-1})\right] \\
&= \prod_{t=1}^{T}|H_t^{-1/2}|\left[\sum_{k=1}^{K}f_{\eta_t(k)}\left(H_t^{-1/2}r_t\right)\mathbb{P}(s_t=k|\Omega_{t-1})\right].
\end{aligned}
$$

Given a sequence of observed returns, the likelihood function of κ can be formulated based on the joint density and estimation can be carried out. We propose a two-step ML approach to estimate the parameter vector κ. Next, we describe the suggested estimator and study its asymptotic properties.

2.3 Two-step ML estimation

When it comes to estimating parameters of regime switching models, Bayesian approaches such as Markov chain Monte-Carlo (MCMC) methods have been mainly used because they are supposed to better deal with path dependence (Hamilton, 2016; Bauwens et al., 2010; Ardia et al., 2018). Ardia et al. (2018) compared the value-at-risk (VaR) forecasting performance of frequentist and Bayesian estimation techniques in a MS-GARCH framework without path dependence. They found that there is no significant performance difference between the rival approaches when it comes to VaR prediction for stock market indices. However, it is important to recognize that the MS-C-MGARCH model is not subject to path dependence, since the conditional second-order process described by the MGARCH model is identical across regimes, and the regime-dependent copula distribution does not incorporate prior regime-dependent information. Thus, the use of ML estimation and Bayesian approaches are both well justified in the present framework. With this being said, despite its asymptotic efficiency, joint estimation of all parameters in the vector $\kappa = (\phi', \psi')'$ in a single step is computationally demanding, given the large dimensionality of the parameter vector. Due to its huge computation cost, the one-step approach becomes hardly operational in finite samples with an increasing number of states and assets in the portfolio. Given these considerations, we suggest a two-step estimator (2SE), which is computationally less expensive (see discussion in Section 2.5 and also Bauwens and Lubrano 1998; Burda 2015). In the first step, the parameter vector ϕ describing the conditional second-order moments is estimated by means of QML. Then, conditional on the first-step estimates, higher-order shape parameters and transition probabilities in ψ are obtained through conditional ML (CML) estimation.

Formally, the two-step estimator reads as $\hat{\kappa}_{T,2SE} = \left(\hat{\phi}'_{T,QML}, \hat{\psi}'_{T,CML}\right)'$. Conditional on a sequence of observed returns $\{r_t\}_{t=1}^{T}$ and some fixed initial values, the first-step QML estimate is the solution of

$\hat{\phi}_{T,QML} = \text{argmax}_{\phi\in\Phi}\, \mathcal{L}_T(\phi)$, where

$$\mathcal{L}_T(\phi) = -\frac{1}{T}\sum_{t=1}^{T} l_t(\phi) = -\frac{1}{T}\sum_{t=1}^{T}\left(r_t' H_t^{-1}(\phi) r_t + \log|H_t(\phi)|\right). \tag{9}$$

Defining $\eta_t(\phi) = H_t^{-1/2}(\phi) r_t$, the second-step estimate is obtained as $\hat{\psi}_{T,CML} = \text{argmax}_{\psi\in\Psi}\, \mathcal{G}_T(\psi, \hat{\phi}_{T,QML})$ with

$$\begin{aligned}\mathcal{G}_T(\psi,\phi) = \frac{1}{T}\log\Bigg[&\sum_{s_1=1}^{K}\cdots\sum_{s_T=1}^{K}\pi^*(s_1)|\Sigma(s_1)^{1/2}|f_{s_1}\left(\Sigma(s_1;\psi)^{1/2}\eta_t(\phi);\psi\right)\\ &\prod_{t=2}^{T} p_{s_t s_{t-1}}|\Sigma(s_t;\psi)^{1/2}|f_{s_t}\left(\Sigma(s_t;\psi)^{1/2}\eta_t(\phi);\psi\right)\Bigg],\end{aligned} \tag{10}$$

where $\pi^*(k)$ is any vector with strictly positive entries s.t. $\sum_{k=1}^{K}\pi^*(k) = 1$.

Regarding algorithmic implementation, we first estimate the MGARCH parameters by means of the Berndt-Hall-Hall-Hausman (BHHH) algorithm (Berndt et al., 1974) with analytical score functions and the Hessian provided by the R-package `BEKKs` (Fülle et al., 2024; Hafner and Herwartz, 2008). Since evaluations of the score and the Hessian are very demanding for MS models, we suggest a numerical optimization algorithm for the second estimation step. More specifically, we employ the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm (Broyden, 1970). Noticing that the likelihood comprises a composition of non-trivial functions, a suitable initial starting value appears crucial to find the global maximum and avoid convergence to local extrema (see OA B for a more detailed algorithm of the two-step estimator).

2.4 Asymptotic properties

In this section, we study the asymptotic properties of the proposed two-step estimator. We outline and discuss sufficient conditions for consistency and asymptotic normality. For simplicity, we denote the dimensions of parameter vectors κ, ϕ, ψ, θ_k, by d_κ, d_ϕ, d_ψ, d_{θ_k}, respectively. Note that $d_\psi = K(K-1) + \sum_{k=1}^{K} d_{\theta_k}$ and $d_\kappa = d_\phi + d_\psi$. We start with three standard regularity conditions on the parameter space and density functions.

Assumption A1. *The true parameter $\kappa_0 \equiv (\phi_0', \psi_0')' \in \mathcal{K}$, where the parameter space $\mathcal{K} \subset \mathbb{R}^{d_\kappa}$ is compact.*

Assumption A2. *For each $k \in \{1,\ldots,K\}$, f_k is absolutely continuous and positive on a neighborhood of 0 with respect to ν^N and $\forall \xi \in \mathbb{R}^N$, $f_k(\xi;\theta_k)$ is continuous on Θ_k.*

Assumption A3. *For all k, $\exists M_1, M_2 < \infty$, s.t., $\rho(\Sigma(k)^{-1}) \leq M_1$, $\mathbb{E}\|\xi_t(k)\|^2 \leq M_2$ and for any $\theta_k \in \Theta_k$, $\mathbb{E}|\log f_k(\xi_t(k);\theta_k)| < \infty$, and there exists an open ball $B(\theta_k,\delta)$, s.t., $\mathbb{E}[\sup_{\tilde{\theta}_k\in B(\theta_k,\delta)} \log^+ f_k(\xi_t(k);\tilde{\theta}_k)] < \infty$.*

The compactness of the parameter space is required for the proof of consistency, although the assumption can be relaxed by adding further requirements on the likelihood function such as concavity. Common compactification techniques can be applied to fulfill Assumption A1. The continuity of density functions f_k leads to the continuity of objective functions as well as identification as we discuss below. Notably, the continuity in Assumption A2 is satisfied if for all k and i, $f_{k,i}$ is absolutely continuous and there exist continuous, strictly decreasing and convex generator functions $\varphi_k : [0,1] \times \Theta_k \mapsto [0,\infty)$ such that the copula admits the representation $C_k(u_1,\ldots,u_N;\theta_k) = \varphi^{-1}\left(\sum_{i=1}^{N} \varphi(u_i;\theta_k);\theta_k\right)$. In other words, Assumption A2 is satisfied, if the copulas under consideration belong to the class of Archimedean copulas, which includes most commonly employed copulas such as Clayton, Frank, Gumbel and product copulas. The integrability conditions in Assumption A3 are required for uniform convergence of the sample objective functions. In addition, the assumption prevents $\xi_t(k)$ from being perfectly correlated. It requires that the lowest eigenvalue of $\Sigma(k)$ is bounded away from zero for all k so that components in the finite mixture (i.e., $\Sigma(k)^{-1/2}\xi_t(k)$) are well-defined and required moments exist. The next assumption requires that the model is identified.

Assumption A4. *For any* $\kappa = (\psi', \phi')' \in \mathcal{K}$, $(\mathcal{L}_T(\phi), \mathcal{G}_T(\psi,\phi))' = (\mathcal{L}_T(\phi_0), \mathcal{G}_T(\psi_0,\phi_0))'$ *implies* $\kappa = \kappa_0$.

The identification of the general class of vec GARCH models, which includes the BEKK specification, was studied by Engle and Kroner (1995). One possible parameter restriction that leads to identification of BEKK(1,1,1) model entails that $\operatorname{diag}(Q), a_{11}, g_{11} > 0$ with a_{11}, g_{11} being the first diagonal element of matrix A and G, respectively. Moreover, the assumption requires that the finite mixture of K components (or less) uniquely determines a unique mixing distribution of η_t (see discussion under Assumption A5). To fix the scale of the distribution of components in the mixture, the marginal variance of elements in random vector $\xi_t(k)$ are normalized to be unity for all k, such that $\Sigma(k)$ is a correlation matrix. Furthermore, the MS model is known to be identified up to a possible permutation of states. The order of the states can be fixed for irreducible chains, which is provided by the next assumption, if the copulas in each states are separable. In addition, even if the same marginal distribution is used across the regimes, the model is identified if C_k's differ for all copula parameters, since copulas and their PDFs are unique for continuous marginals under the previous assumption (Sklar, 1959). For this reason, we exclude the independence case for at least $K-1$ copula parameter spaces and assume distinct copulas for all states. Following Bauwens et al. (2010), we suggest starting far from independence-implying copula parameters to avoid getting into states that lack clear separability. The next two assumptions jointly establish the strict stationarity and ergodicity of the observable sequence $\{r_t\}$.

Assumption A5. *The transition probability matrix P is ergodic.*

Assumption A5 implies that the chain $\{s_t\}$ is irreducible and aperiodic. By the Perron-Frobenius Theorem,

$\{s_t\}$ converges exponentially fast to a unique stationary distribution $\pi(k)$ regardless of the initial distribution with $\pi(k) > 0$ for all k. Moreover, the ergodicity extends to the sequence $\{\eta_t\}$ by Lemma 1 of Leroux (1992). It is evident that the marginal distribution of the MGARCH innovation is a finite mixture from the parametric family of K copula distributions, i.e.,

$$\eta_t = \sum_{k=1}^{K} w_k \eta_t(k),$$

where the weights w_k are Bernoulli random variables with success probability given by the stationary distribution of the chain, i.e., $\mathbb{P}(w_k = 1) = \pi(k)$. Therefore, $\{\eta_t\}$ is an ergodic strictly stationary martingale difference sequence with marginal density (with respect to ν^N) given by $f_\eta = \sum_{k=1}^{K} \pi(k) f_{\eta_t(k)}$.

To establish the stationarity and geometric ergodicity of the MGARCH process, note that the conditional covariance can be embedded in a stochastic recurrent equation (see, e.g. Straumann and Mikosch, 2006):

$$\text{vech}\,(H_t) = \boldsymbol{A}_{t-1}\text{vech}\,(H_{t-1}) + \boldsymbol{B}_t, \tag{11}$$

where $\boldsymbol{B}_t = D_N^+(Q \otimes Q)\text{vec}\,(I_N)$ with D_N being the duplication matrix of dimension $N^2 \times \frac{N(N+1)}{2}$ and D_N^+ its Moore-Penrose inverse and $\boldsymbol{A}_{t-1} = D_N^+ \left((A \otimes A)'\Delta_{t-1}(\phi, \psi) + (G \otimes G)'\right) D_N$ with

$$\Delta_t(\phi, \psi) = N_N(H_t^{1/2}\eta_t \otimes I_N)(\eta_t' \otimes I_N)(N_N(H_t^{1/2} \otimes I_N))^{-1},$$

where N_N is a symmetric idempotent matrix such that $N_N = \frac{1}{2}(I_{N^2} + K_{NN})$ with K_{NN} being the commutation matrix (see Magnus and Neudecker, 1999, for application of duplication and commutation matrices). It is well-known in the literature on stochastic dynamic systems that the process in (11) has a nonanticipative strictly stationary solution if the associated top Lyapunov exponent is strictly negative (see Bougerol and Picard 1992 and Ch.2.2.2 in Francq and Zakoian 2019):

$$\gamma = \inf\left\{\mathbb{E}\left[\frac{1}{n+1}\log\|\boldsymbol{A}_0\boldsymbol{A}_{-1}\ldots\boldsymbol{A}_{-n}\|\right],\ n \in \mathbb{N}\right\} < 0. \tag{12}$$

The sufficient condition for $\gamma < 0$ is stated in the following assumption (see proofs in Appendix 5 for details).

Assumption A6. $\sup_{\kappa\in\mathcal{K}} \mathbb{E}\left[\log\|D_N^+\left((A \otimes A)'\Delta_t(\phi, \psi) + (G \otimes G)'\right) D_N\|\right] < 0.$

Note that if one chooses a symmetric matrix square root $H_t^{1/2}$, the commutation matrix simplifies to $K_{NN} \equiv \frac{\partial \text{vec}\left(H_t^{1/2}\right)}{\text{vec}\left(H_t^{1/2}\right)'} = I_{N^2}$ and $N_N = I_{N^2}$, thus $\Delta_t(\phi, \psi) = (H_t^{1/2}\eta_t\eta_t'H_t^{-1/2} \otimes I_N)$, which was considered by Hafner and Preminger (2009). Assumption A6 can also be replaced by a sufficient condition:

$$\rho\left(D_N^+((A \otimes A)' + (G \otimes G)')D_N\right) < 1, \tag{13}$$

which will also lead to second-order stationarity (see Comte and Lieberman, 2003; Boussama et al., 2011). Let $\tilde{H}_t(\phi) = H(\eta_t, \ldots, \tilde{\eta}_0, \tilde{\eta}_{-1}; \phi)$ be the conditional covariance process where the initial values are drawn from the stationary distribution. Let $\tilde{l}_t(\phi)$ and $\tilde{\mathcal{L}}_T(\phi)$ be defined analogously. The next assumption establishes the so-called invertibility condition such that initial values are asymptotically irrelevant and $\tilde{\mathcal{L}}_T(\phi)$ and $\mathcal{L}_T(\phi)$ converge to the same limit.

Assumption A7. *$\rho(G \otimes G) < 1$ and $\mathbb{E}\|\tilde{H}_t(\phi_0)\|^s < \infty$ for some $s > 0$.*

Theorem 1. *Under Assumptions (A1) – (A7),*

$$\hat{\kappa}_{T,2SE} \to \kappa_0, \ a.s., \text{ as } T \to \infty.$$

Proof. A proof is provided in Appendix 5 □

To establish asymptotic normality, we make the following assumptions.

Assumption A8. *The true parameter $\kappa_0 \in$ interior$(\mathcal{K})$, $\mathbb{E}\|r_t\|^6 < \infty$ and for s in Assumption A7, $s > 1$.*

Assumption A9. *For each $k \in \{1, \ldots, K\}$, f_k is twice continuously differentiable on a open neighborhood $B(\theta_{0,k}, \delta)$ of $\theta_{0,k}$ and for all $i, j \in \{1, \ldots, d_{\theta_k}\}$,*
(i) $\int \sup_{\theta_k \in B(\theta_{0,k},\delta)} |\partial f_k(\xi;\theta_k)/\partial\theta_{k,i}| \nu^N(d\xi) < \infty$, $\int \sup_{\theta_k \in B(\theta_{0,k},\delta)} |\partial^2 f_k(\xi;\theta_k)/\partial\theta_{k,i}\partial\theta_{k,j}| \nu^N(d\xi) < \infty$;
(ii) $\mathbb{E}[\sup_{\theta_k \in B(\theta_{0,k},\delta)} |\partial \log f_k(\xi_t(k);\theta_k)/\partial\theta_{k,i}|^2] < \infty$, $\mathbb{E}[\sup_{\theta_k \in B(\theta_{0,k},\delta)} |\partial^2 \log f_k(\xi_t(k);\theta_k)/\partial\theta_{k,i}\partial\theta_{k,j}|] < \infty$;
(iii) for any $k, l, s_1 \in \{1, \ldots, K\}$, $\mathbb{P}(\sup_{\theta_k, \theta_l \in B(\theta_{0,k},\delta) \cap B(\theta_{0,l},\delta)} \sup_{k,l} f_k(\xi_1;\theta_k)/f_l(\xi_1;\theta_l) = \infty | s_1) \neq 1$.

We exclude the case in which the true parameter lies on the boundary of the parameter space. The moment condition in Assumption A8 is required for the finiteness of the information matrix. Regularity conditions outlined in Assumption A9 include sufficient conditions that (i) permit interchanging differentiation with integration, (ii) existence of the respective dominating functions required for convergence of sample moments, and (iii) a condition preventing densities in different states from having disjoint support. The next Theorem summarizes the asymptotic distribution of the proposed estimator.

Theorem 2. *Under Assumptions (A1) – (A9),*

$$\sqrt{T}\left(\hat{\kappa}_{2SE} - \kappa_0\right) \xrightarrow{d} \mathcal{N}\left(0, \begin{pmatrix} \mathcal{M}_{1,1} & \mathcal{M}_{1,2} \\ \mathcal{M}_{2,1} & \mathcal{M}_{2,2} \end{pmatrix}\right),$$

where the exact expressions for blocks $\mathcal{M}_{i,j}$ are documented in Appendix 5.

Proof. A proof is provided in Appendix 5 □

2.5 Extensions towards higher dimensions

The empirical analysis provided in this work has focused on the case of bivariate return processes ($N = 2$). When moving towards higher dimensions, the model implementation becomes more intricate for (at least) two reasons. On the one hand, the number of parameters of the MGARCH part of the model (i.e., the BEKK specification) increases at a rate of $\mathcal{O}(N^2)$ and invokes a curse of dimensionality. On the other hand, modeling the dependence structure by means of (standardized) Vine copula distributions becomes increasingly complicated in higher dimensions, since the number of parameters also increases by $\mathcal{O}(N^2)$. Regarding the former issue, the unrestricted BEKK(1,1,1) model in (3) may be simplified to a diagonal BEKK (Engle and Kroner, 1995), where the parameter matrices A and G are restricted to be diagonal, or even further to a model specification with scalar parameters as suggested by Ding and Engle (2001). However, such approaches reduce the model flexibility for addressing volatility spillovers. Against this background, it is worth noting that the recently developed R package `BEKKs` (Fülle et al., 2022, 2024) allows fast estimation of full BEKK(1,1,1) models, such that - for instance - model estimation for $T = 5,000,\ N = 5$ is accomplished in less than one minute. Regarding the second source of model complexity, two strategies are worth considering. First, as described in Nelsen (2006) and Lee and Long (2009), one might opt for single parameter Archimedean copulas in higher dimensions. However, an apparent disadvantage of such a parsimonious approach is that all pairwise dependencies are homogeneous by assumption. Second, keeping the flexibility of heterogeneous bivariate dependencies, Vine copulas can be simplified by assuming independence of some bivariate relations, and a focus on (economically) most relevant dependencies. As an empirical 'test of concept' Fülle and Herwartz (2024) employ the suggested MS-C-MGARCH model for the ex-ante assessment of tail risks of portfolios that consist of three constituent asset classes namely equity, gold, and treasury bond futures. In this higher dimensional examination, the MS-C-MGARCH model outperforms benchmark volatility models (MGARCH, C-MGARCH) in measuring one-day-ahead value-at-risk and expected shortfall. Moreover, the performance lead of the three-state MS-C-MGARCH model is stronger, if the share of risky assets (i.e., equity) is relatively large.

Unlike the description of covariance profiles or higher-order dependencies, the parameterization of MS patterns is not subject to the dimensionality of the return process. However, the parameters of the transition matrix M increase with the number of states K as $\mathcal{O}(K^2)$. It is therefore important to keep the number of states reasonably small and avoid superfluous states. We introduced MS by means of regime-specific copulas in Section 2.2. Apart from regime-specific dependencies, a possible model extension could be to account for changing marginal distributions beyond the baseline specification (6). For instance, one could think of having left-skewed and (or) leptokurtic marginals in times of economic crisis and right-skewed marginals for

periods of recovery to address stylized facts of financial time series (Singleton and Wingender, 1986; Harvey and Siddique, 1999; Peiró, 1999; Bai et al., 2003; Carnero et al., 2004). Flexible candidate distributions to account for such patterns are the skewed t-distribution (Jones and Faddy, 2003) or the skewed-normal (Azzalini, 1985) distribution. Moreover, these marginals can be combined with different copula regimes to additionally account for changing dependencies. However, the admission of state-specific marginals might easily suffer from a curse of dimensionality.

3 Monte-Carlo analysis

This section examines the finite-sample properties of the two-step estimator via Monte-Carlo experiments.

3.1 Data generation and performance assessment

Following previous studies, we employ a BEKK(1,1,1) model with parameters ϕ calibrated as (see also Lütkepohl, 2005; Hafner and Herwartz, 2008)

$$Q = \begin{pmatrix} 0.50 & 0 \\ -0.05 & 0.50 \end{pmatrix} \; A = \begin{pmatrix} 0.30 & -0.20 \\ 0.10 & 0.30 \end{pmatrix}, \text{ and } G = \begin{pmatrix} 0.91 & -0.02 \\ 0.03 & 0.90 \end{pmatrix}.$$

The parameters fulfill the stationarity condition (13) with a maximal eigenvalue of 0.93 in modulus. For a thorough assessment of the estimation performance, a variate of parameters and distributions are used. Specifically, we set up three alternative scenarios for the two regime distributions with distinct parameter specifications:

1. I/G: Independence ($s_t = 1$) combined with Gumbel copula ($s_t = 2$),
2. C/G: Clayton ($s_t = 1$) combined with Gumbel copula ($s_t = 2$),
3. C/RG: Clayton ($s_t = 1$) combined with 90°-rotated Gumbel copula ($s_t = 2$).

To capture alternating dependencies and Markov processes, we set up different parameter sets within those distributional settings. First, we construct a DGP with persistent states, i.e., large Markov probabilities and at least one state with high dependency ($p_{11} > 0.9$ and $p_{22} > 0.9$; $\theta_c > 2$). Second, we set up a case with one predominant regime and moderate dependencies (either $p_{11} < 0.5$ or $p_{22} < 0.5$; $\theta_c < 2$).

In each simulation experiment, given the BEKK(1,1,1)-parameters ϕ, and selected copulas C_1, C_2 – with respective parameters $\theta_{1,c}, \theta_{2,c}$ and transition probabilities p_{11} and p_{22} – a bivariate time series with length T is drawn according to the following procedure:

1. Set the initial covariance matrix of the BEKK(1,1,1) process, i.e., H_1, to the unconditional covariance matrix: $H_1 = (I_4 - (A \otimes A)' - (G \otimes G)')^{-1} \cdot \text{vec}(QQ')$.

2. Estimate the covariance matrix $\Sigma(k)$ for the two copula distributions from two iid samples of 100,000 observations of C_k with parameter $\theta_{k,c}$ and compute $\Sigma(k)^{-1/2}$.

3. Sample the initial distribution by means of a Bernoulli trial (returning 1 or 2) where the probability of the two states is identical. Then sample $\xi_1(s_1)$ and determine $e_1 = H_1^{1/2}\Sigma(s_1)^{-1/2}\xi_1$.

4. Iterate through the following steps for $t = 2$ until $t = T + 200$:

   (a) Perform a Bernoulli trial (returning $s_t = 1$ or $s_t = 2$), where the probability of returning $s_t = 1$ is dependent on the previous state, i.e., $\mathbb{P}(s_t = 1|s_{t-1} = 2) = 1 - p_{22}$ and $\mathbb{P}(s_t = 1|s_{t-1} = 1) = p_{11}$.

   (b) If $s_t = k$: Generate a $C_k(\cdot, \cdot; \theta_{k,c})$-distributed random variable ξ_t and determine $\eta_t = \Sigma(k)^{-1/2}\xi_t$ for $k \in \{1, 2\}$.

   (c) Determine H_t as outlined in (3) and $e_t = H_t^{1/2}\eta_t$.

5. Discard the first 200 values of the generated series and subject the simulated process to ML estimation.

For the simulation study $R = 1,000$ repetitions of random series with $T = 1,000$, 2,500, 5,000 and 10,000 observations are generated for each DGP specification. To avaluate the estimation performance, we document the following descriptive statistics:

$$\text{mean}(\hat{\kappa}) = \overline{\hat{\kappa}} := \frac{1}{R}\sum_{r=1}^{R}\hat{\kappa}_r, \quad \text{sd}(\hat{\kappa}) := \sqrt{\frac{1}{R-1}\sum_{r=1}^{R}\left(\hat{\kappa}_r - \overline{\hat{\kappa}}\right)^2}, \quad \text{MSE}(\hat{\kappa}) := \frac{1}{R}\sum_{r=1}^{R}\left(\hat{\kappa}_r - \kappa_0\right)^2,$$

with $(\hat{\kappa}_r)_{r=1,\ldots,R}$ being the sequence of estimates from R Monte-Carlo experiments. In addition, we assess the accuracy of state detection by documenting the hit rate for estimated state variables $\hat{s}_t$ defined by

$$h(\hat{s}) = \frac{1}{T}\sum_{t=1}^{T}(s_t - 1)\mathbb{I}\left(\mathbb{P}(\hat{s}_t = 1|\Omega_t) < 0.5\right) + (2 - s_t)\mathbb{I}\left(\mathbb{P}(\hat{s}_t = 1|\Omega_t) \geq 0.5\right),$$

where $\mathbb{I}(\cdot)$ is the indicator function returning a value of one if the condition is fulfilled and zero otherwise.

3.2 Simulation results

Simulation results, including descriptive statistics and estimated kernel densities, for the second step of the estimation, conditional on true BEKK(1,1,1) estimators, are presented in Table 1 below and in Figures C.1 and C.2 in OA C, respectively. Additional results, where the true BEKK parameters are replaced by QML

		Independence and Gumbel (I/G)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.93	0.749	0.307	0.127	0.833	0.226	0.060	0.885	0.147	0.024	0.910	0.087	8.0e-3
p_{22}	0.965	0.880	0.178	0.039	0.916	0.123	0.017	0.944	0.073	5.8e-3	0.955	0.043	2.0e-3
θ_G	2.3	2.358	0.357	0.131	2.326	0.185	0.035	2.307	0.121	0.015	2.309	0.089	8.0e-3
$h(\hat{s})$			0.672			0.684			0.689			0.691	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.15	0.219	0.258	0.071	0.235	0.273	0.082	0.229	0.267	0.077	0.240	0.259	0.075
p_{22}	0.98	0.907	0.177	0.037	0.946	0.102	0.011	0.957	0.065	4.8e-3	0.968	0.049	2.6e-3
θ_G	1.35	1.385	0.110	0.013	1.363	0.062	4.0e-3	1.361	0.044	2.0e-3	1.353	0.031	9.6e-4
$h(\hat{s})$			0.956			0.974			0.976			0.977	
		Clayton and Gumbel (C/G)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.92	0.895	0.081	7.2e-3	0.913	0.024	6.5e-4	0.917	0.016	2.5e-4	0.918	0.011	1.3e-4
p_{22}	0.97	0.962	0.027	7.8e-4	0.967	0.010	1.1e-4	0.969	0.007	5.0e-5	0.969	0.005	2.1e-5
θ_C	4	4.038	0.654	0.429	4.025	0.376	0.142	4.013	0.276	0.076	4.031	0.189	0.037
θ_G	1.3	1.314	0.109	0.012	1.307	0.068	4.6e-3	1.301	0.044	1.9e-3	1.301	0.031	9.8e-4
$h(\hat{s})$			0.798			0.799			0.802			0.802	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.38	0.395	0.294	0.087	0.363	0.240	0.058	0.364	0.192	0.037	0.372	0.150	0.023
p_{22}	0.75	0.742	0.164	0.027	0.735	0.119	0.014	0.743	0.085	7.3e-3	0.748	0.063	4.0e-3
θ_C	1.2	1.274	0.579	0.341	1.223	0.276	0.076	1.207	0.167	0.028	1.208	0.122	0.015
θ_G	1.6	1.617	0.176	0.031	1.603	0.091	8.3e-3	1.605	0.064	4.1e-3	1.600	0.046	2.2e-3
$h(\hat{s})$			0.703			0.716			0.718			0.720	
		Clayton and 90° Gumbel (C/RG)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.96	0.950	0.036	1.4e-3	0.955	0.014	2.3e-4	0.958	9.0e-3	9.3e-5	0.958	6.0e-3	4.1e-5
p_{22}	0.915	0.893	0.075	6.6e-3	0.906	0.030	9.9e-4	0.910	0.019	3.8e-4	0.912	0.012	1.6e-4
θ_C	2.3	2.306	0.209	0.044	2.295	0.130	0.019	2.302	0.091	8.2e-3	2.293	0.065	4.3e-3
θ_{G90}	-3.53	-3.812	1.345	1.886	-3.631	0.580	0.346	-3.589	0.376	0.145	-3.592	0.258	0.070
$h(\hat{s})$			0.766			0.768			0.767			0.768	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.27	0.308	0.280	0.080	0.297	0.234	0.055	0.277	0.184	0.034	0.255	0.150	0.023
p_{22}	0.85	0.839	0.086	7.5e-3	0.832	0.060	3.7e-3	0.842	0.047	2.2e-3	0.84	0.035	1.4e-3
θ_C	1.13	1.191	0.574	0.333	1.154	0.379	0.144	1.154	0.229	0.053	1.147	0.155	0.024
θ_{G90}	-6.3	-6.566	1.756	3.216	-6.407	0.954	0.951	-6.31	0.685	0.481	-6.264	0.441	0.199
$h(\hat{s})$			0.829			0.832			0.833			0.833	

Table 1: Monte-Carlo results for the MS and copula parameters assuming true BEKK(1,1,1) parameters (mean, standard deviation and MSE of parameter estimates and the frequencies $h(\hat{s})$ of correct state assessment). The number of Monte-Carlo repetitions is 1,000.

estimates, are provided in Table C.1 and Figur C.3 in OA C. The results of the Monte-Carlo simulation offer insights into the performance of the proposed estimator, highlighting several key aspects of its behavior:

1. All generated processes can be consistently estimated. For instance, the simulation results for the case Clayton combined with Gumbel (C/G) and parameters ($p_{11} = 0.92, p_{22} = 0.97, \theta_C = 4, \theta_G = 1.3$) show

that average estimates approach the true parameters and the respective MSE statistics decrease with expanding sample sizes. To visualize these results, Figures C.1 and C.2 in OA C display kernel density estimates. The consistency broadly holds for all considered DGPs.

2. Finite-sample performance depends on depends on the accuracy of state assignment. More persistent states (p_{11} and/or p_{22} values close to 1) and clearly distinguishable copula shapes lead to smaller standard deviations and MSE statistics. For instance, the C/G case shows that even if the standard deviation is shrinking in larger samples, the descriptive statistics highlight more pronounced finite-sample biases and standard deviations in the presence of relatively less persistent regimes. For instance, the MSE is larger for all parameters in the C/G simulation design when $p_{11} = 0.92$ reduces to $p_{11} = 0.38$ and $p_{22} = 0.97$ reduces to $p_{22} = 0.75$. Hence, it seems that with at least one small probability of staying in a regime, i.e., $p_{ii} < 0.5$, higher MSE statistics are obtained especially for the comparatively small transition probability parameter (see, for instance, the independence combined with Gumbel (I/G) setup where for $p_{11} = 0.15$, and the MSE, standard deviation and bias are significantly larger than for the setup with $p_{11} = 0.93$).

3. The estimates of transition probability parameters p_{11} and p_{22} are less biased and have a lower MSE than the copula parameter estimates in all simulation designs. Accordingly, the simulation results indicate that estimating the correct copula parameters is more challenging in finite samples compared to the assessment of state transition probabilities.

4. The detection of the states improves with larger sample sizes. However, even if the parameters are close to the true probabilities on average, the detection of the correct states is demanding. The large hit rate in the I/G case with one predominant regime seems to signify a comparably accurate state estimation. Nevertheless, the frequent detection of correct states grounds in one persistent state that is due to the specific MS parameters of $p_{11} = 0.15$ and $p_{22} = 0.98$. Moreover, a fraction of the overall deviation from the correct states is likely due to the switching mechanism. Even with a high probability of being in one state, the random draw could still end up in the other state, leading to an incorrect representation of s_t.

5. Estimating the MS and the copula parameters yields similar results for the two-step estimator and a practically infeasible alternative for the second estimation step that conditions on the true BEKK parameters. This can be seen from a comparative assessment of estimated kernel densities in Figures C.2 and C.3 in OA C, or of Table 1 and Table C.1 in OA C. Hence, consistency of the MS and copula parameters is unaffected by first-step estimation of the BEKK parameters.

In summary, the Monte-Carlo results underscore the strengths of the proposed two-step estimator, as the performance remains largely unchanged whether using the first-step estimated BEKK parameters or their true values. Additionally, the MS-C-MGARCH model effectively captures flexible dependence structures, demonstrating its ability to detect even mild changes in dependence patterns with sufficient sample information.

4 Hedge or hazard? Evaluating Bitcoin's role during stock market turmoil

In this section, we provide an empirical illustration of the proposed MS-C-MGARCH framework and shed light on its advantages by comparing it with two benchmark models including time-invariant C-MGARCH models of Lee and Long (2009) and a classical Gaussian MGARCH model. We apply the new model and its benchmarks to a bivariate time series consisting of daily log-returns of Bitcoin and the NASDAQ Composite index (hereinafter: BTC and NASDAQ, respectively). For simplicity, we deploy a BEKK(1,1,1) specification to evaluate time-varying covariances that apply to all alternative models (i.e., Gaussian BEKK, C-BEKK and MS-C-BEKK). We start by providing a statistical assessment of modeling performance in terms of goodness-of-fit and according to information criteria. In the light of the recent development of digital currencies, we subsequently examine whether BTC holds potential to act as a hedge in times of stock market turmoil and economic downturns. Comprising 6825 log-return observations, the sample covers the period from September 18, 2014 until December 31, 2023. The analyzed log-returns are displayed in Figure D.4 in OA D. Evidently, both processes are subject to (partly synchronized) volatility clustering. Some degree of positive comovements is also visible. The unconditional correlation between the two return series is moderate, i.e., 0.23.

Parameter estimates are shown in Table 2, whereas model-implied conditional standard deviations of the return processes and correlations are displayed in Figure 1. Visual inspection of model-implied standard deviations suggests that the return of BTC is generally more volatile compared to that of the NASDAQ index over the entire sample period. The conditional correlation pattern unveils that BTC and NASDAQ were largely uncorrelated in the pre-pandemic era. A strong positive correlation emerged during the global outbreak of the COVID-19 virus in 2020 but vanished in the course of 2023. While there is also some mild evidence in favor of negative conditional correlations (for instance in 2019), the observations of mostly absent or positive correlations strongly limit the scope of hedging NASDAQ exposure using BTC. Unlike linear correlations, however, using the MS-C-MGARCH framework to detect higher-order relationships as well as their time variation can provide valuable insights if holding BTC is subject to excess portfolio loss in periods

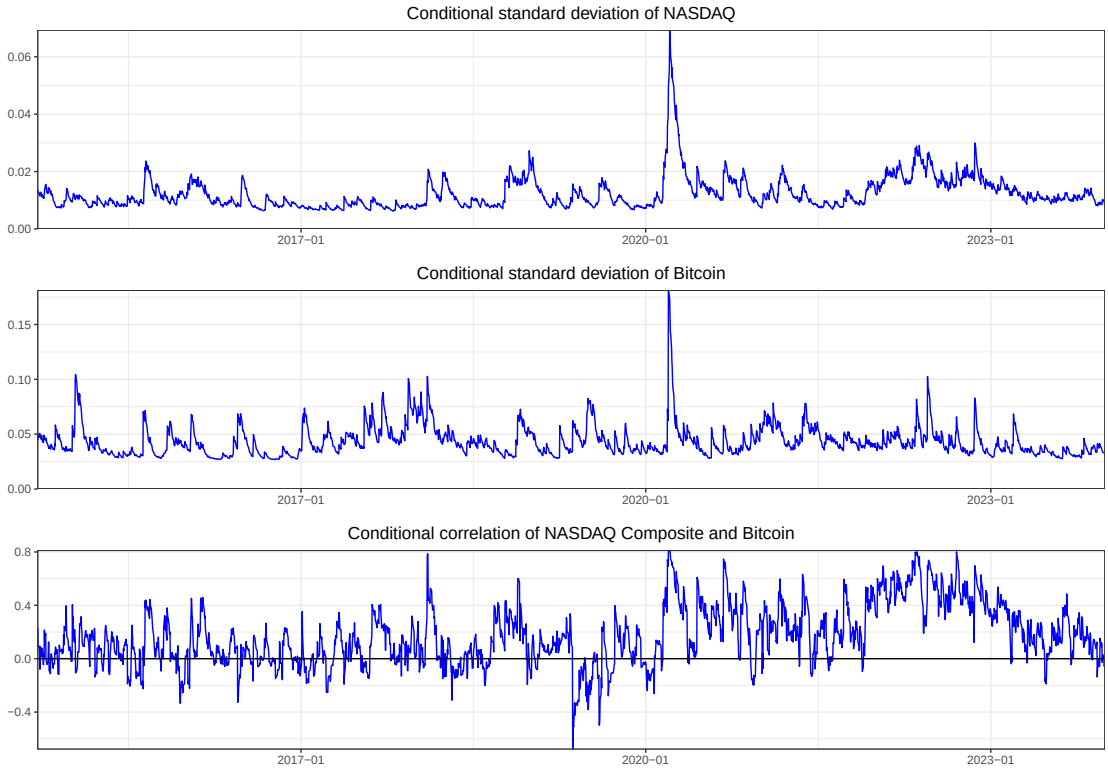


Figure 1: Estimated conditional correlations and standard deviations of the bivariate return series. Estimation results and plot origin from the R-package `BEKKs` Fülle et al. (2022).

$$\widehat{Q} = \begin{pmatrix} \text{2.17e-3} & 0 \\ (\text{4.82e-4}) & \\ \text{7.52e-4} & \text{9.99e-3} \\ (\text{2.86e-3}) & (\text{2.36e-3}) \end{pmatrix}, \quad \widehat{A} = \begin{pmatrix} 0.356 & 0.284 \\ (0.051) & (0.300) \\ \text{-2.59e-3} & 0.284 \\ (\text{6.68e-3}) & (0.055) \end{pmatrix}, \quad \widehat{G} = \begin{pmatrix} 0.921 & \text{-0.073} \\ (0.023) & (0.130) \\ \text{9.58e-4} & 0.923 \\ (\text{3.37e-3}) & (0.025) \end{pmatrix}.$$

Table 2: BEKK(1,1,1) parameter estimates with QML standard errors in parentheses.

of market downturn.

Regarding higher-order dependence structures, we follow Lee and Long (2009) and focus on prominent Archimedean copulas with Gaussian marginals, i.e., Gumbel (G), Gumbel survival (GS), Clayton (C) and Clayton survival (CS; see OA A for definitions of the respective copula functions and parametrization). We next discuss model selection and estimation results for alternative model implementations (MGARCH, C-MGARCH, MS-C-MGARCH). Conditional on a particular best-performing model and going beyond return correlations, we subsequently investigate the prevalence of higher-order dependence structures from September 2019 until June 2020, in particular, the surge marking the outbreak of the COVID crisis in March 2020.

4.1 Comparative performance of alternative models

For model comparisons we consult (i) values of log-likelihoods measured at the respective parameter estimates (llv), (ii) the Akaike information criterion, AIC $= -2 \cdot \text{llv} + 2 \cdot J$ with J being the number of model parameters, and (iii) the Bayesian information criterion, BIC $= -2 \cdot \text{llv} + J \cdot \ln(T)$. The C-MGARCH model nests the conditionally Gaussian model. Hence, explicit likelihood ratio (LR) tests can also be used to evaluate, if the more general model class improves goodness-of-fit after penalizing the augmentation of the parameter space.

Estimation results comparing the Gaussian MGARCH model with various C-MGARCH specifications are presented in Table 3. Unlike using distributional models implying lower-tail dependencies (Clayton or Gumbel survival copula), specifying the C-MGARCH model with a Clayton survival copula (llv = 11360.5) or a Gumbel copula (llv = 11350.6) fails to improve goodness-of-fit achieved by the Gaussian BEKK model (llv = 11350.8). According to both model selection criteria, the C-MGARCH model with the Clayton copula shows up as the preferable specification. While a well-specified C-MGARCH model can offer moderate improvements in goodness-of-fit, this comes with certain trade-offs. Firstly, C-MGARCH models introduce issues of copula selection and require the estimation of additional parameters compared with the Gaussian MGARCH model, which assumes conditional independence. Secondly, the presence of nuisance parameters may complicate the model's interpretability. As Lee and Long (2009) pointed out in footnote 12, the complexity in interpreting C-MGARCH models arises from how tail dependence in the model innovations, ξ_t, is transmitted to higher-order dependencies in the return residuals, e_t. Specifically, this transmission is likely subject to nuisance parameters given (i) the multivariate standardization that orthogonalizes copula random variables ($\eta_t = \Sigma^{-1/2}\xi_t$), and (ii) the subsequent scaling of these variables to the MGARCH-implied covariances ($e_t = H_t^{1/2}\eta_t$). Noticing that these challenges can affect both the model feasibility and interpretability, one might argue that the motivation for adopting a C-MGARCH framework hinges on its ability to enhance goodness-of-fit.[3] In light of these concerns, we now turn to our novel framework: a regime-switching copula-MGARCH model.

Estimation results for two-state MS-C-MGARCH models ($K = 2$) are shown in Table 4. All two-state MS-C-MGARCH specifications outperform the best-fitting C-MGARCH model in terms of goodness-of-fit, even after accounting for the additional parameters. The estimated copula parameters indicate strong (asymmetric) dependencies, and the changes in these dependence parameters compared to the C-MGARCH model demonstrate the model's ability to adapt to shifting financial market conditions. These dynamics, which may go unnoticed in time-invariant C-MGARCH models that only capture mild higher-order dependencies, underscore the value of introducing regime-switching mechanisms. Among the different asymmetric dependence

[3] To elicit the transmission in question for a given exemplary case, Fülle and Herwartz (2022) suggest finite sample Monte-Carlo analysis. In such a controlled environment 'returns' (i.e., return residuals e_t) are drawn from an estimated MGARCH model processing standardized copula distributed innovations that are subject to Markov switching with estimated transition probabilities. Both the (state-dependent) copula functions and their parameterization have to be chosen by the analyst to grasp the full flexibility of the transmission from model innovations ξ_t to return residuals e_t.

Copulas	Estimates	Goodness-of-fit criteria		
C	$\hat{\theta}$	llv	AIC	BIC
I	-	11350.8	-22733.2	-22810.6
G	1.000 (0.036)	11350.6	-22725.09	-22802.5
GS	1.016 (0.010)	11354.6	-22733.2	-22810.6
C	0.060 (0.020)	**11356.9**	**-22737.8**	**-22815.2**
CS	0.000 (0.008)	11350.5	-22725.08	-22802.5

Table 3: Estimation results and goodness-of-fit criteria for alternative C-MGARCH models with a BEKK(1,1,1) conditional (co)variance specification as documented in Table 2. We use the following abbreviations for copula functions: Clayton (C), Clayton survival (CS), Gumbel (G), Gumbel survival (GS) and Independence (I). The largest llv and the smallest AIC and BIC statistics are in boldface.

Copula	Estimates			Information criteria		
C	$\hat{p}_{11}$	$\hat{p}_{22}$	$\hat{\theta}_C$	llv	AIC	BIC
I /G	0.133 (0.138)	0.409 (0.107)	29.868 (19.544)	**11502.6**	**-22977.2**	**-22886.9**
I /GS	0.399 (0.137)	0.071 (0.176)	11.166 (4.608)	11423.6	-22819.2	-22728.9
I /C	0.810 (0.137)	0.153 (0.176)	23.165 (4.608)	11410.1	-22792.2	-22701.9
I /CS	0.684 (0.229)	0.153 (0.369)	19.773 (9.167)	11465.8	-22903.5	-22813.2

Table 4: Parameter estimation and information criteria for alternative two-state MS-C-MGARCH models combining a specific copula distribution with an independence state (I). Estimated standard errors in parentheses are calculated as outlined in Section 2.4. For further notes see Table 3.

structures under consideration, the combination of an independence state with a Gumbel copula emerges as the best-fitting model, effectively capturing upper-tail dependence between BTC and NASDAQ. In contrast, the second-best model features a lower-tail dependence between BTC and NASDAQ returns. The relatively small to moderate probabilities of remaining in either regime (independence or copula) suggest frequent transitions between dependence patterns, highlighting the model's ability to capture dynamic shifts in market behavior.

Given the evidence of distinct dependence structures, it is worthwhile to explore whether combining multiple higher-order dependence profiles through three-state MS-C-MGARCH models ($K = 3$) can further enhance model performance. The results for the three-state models are shown in Table 5. This analysis focuses on copula combinations featuring both upper- and lower-tail dependencies. The best-performing three-state specification includes an independence state alongside Gumbel and Clayton copulas, and it remarkably outperforms the two-state model, as indicated by an LR statistic of 125.02. Although the exact distribution

Copulas	Estimates								Information criteria		
$C_1/C_2/C_3$	$\hat{p}_{11}$	$\hat{p}_{12}$	$\hat{p}_{21}$	$\hat{p}_{22}$	$\hat{p}_{32}$	$\hat{p}_{33}$	$\hat{\theta}_1$	$\hat{\theta}_2$	llv	AIC	BIC
I /G /GS	0.563	0.174	0.000	0.784	0.298	0.472	18.770	30.287	11561.2	-23084.5	-22961.9
	(0.057)	(0.221)	(0.082)	(0.005)	(0.093)	(0.172)	(5.734)	(20.806)			
I/G/C	0.813	0.122	0.022	0.823	0.576	0.424	18.773	30.286	**11565.1**	**-23092.2**	**-22969.7**
	(0.169)	(0.114)	(0.043)	(0.052)	(0.175)	(0.078)	(10.489)	(19.042)			
I/GS/CS	0.684	0.315	0.053	0.565	0.423	0.411	26.565	31.433	11556.0	-23074.0	-22951.5
	(0.053)	(0.867)	(0.289)	(0.624)	(0.520)	(0.225)	(47.453)	(11.631)			
I/C/CS	0.776	0.224	0.086	0.199	0.440	0.394	11.747	13.276	11537.5	-23037.0	-22914.5
	(0.119)	(0.236)	(0.136)	(0.169)	(0.131)	(0.116)	(1.968)	(1.693)			

Table 5: Estimation results for the three-state MS-C-MGARCH model. Here, C_i stands for the copula of state i. For further notes see Table 3.

of the LR statistic is nonstandard in the present MS framework, the improvement is likely to be substantial, given that the three-state model only requires five additional parameters in comparison with the two-state model. Thus, the data strongly suggest deviations from the Gaussian model (i.e., the independence copula) and indicate the presence of regime switches between heterogeneous higher-order dependence structures. Table 5 also provides the estimated transition probabilities. Notably, in the best-performing model, estimated probabilities for remaining in states of independence and upper-tail dependence (Gumbel copula) are above 0.8 to indicate considerable persistence in these regimes. By contrast, the Clayton state (lower-tail dependence) exhibits a relatively lower staying probability of 0.42. Moreover, the data suggest that transitions into the independence or upper-tail dependence states are most likely to originate from similar states, while transitions into the lower-tail dependence state (Clayton copula) are unlikely to occur from the independence state.

Considering the estimated state probabilities, the lack of a single dominant state raises the possibility that a static mixture of copulas could yield goodness-of-fit results that are comparable to those of the MS-C-MGARCH model. However, mixing the independence copula with dependence states does not call the proposed MS-C-MGARCH model into question. Since such combinations of PDFs imply (mild) dependencies across the series, mixing the independence copula with dependence states aligns well with findings of Lee and Long (2009); Erb et al. (1994); Longin and Solnik (2001); Ang and Chen (2002) and Hsu et al. (2008). Given that tail dependence patterns tend to evolve over time, the estimated state probabilities may reflect time-varying 'relative' weights of different market conditions, suggesting that mixture distributions could enhance our understanding of complex financial market processes. Overall, our findings underscore the significance of asymmetric dependencies and the presence of structural changes in such dependencies over time.

4.2 Changing dependence structures in the wake of the pandemic

To uncover the time-varying nature of model-implied dependencies, we examine the smoothed state probabilities $\mathbb{P}(s_t = i|\Omega_T)$. Figure 2 presents the empirical returns, conditional correlations, and estimated state

probabilities for the period from September 2019 to June 2020. We have already gained the impression from Figure 1 that conditional correlations between BTC and NASDAQ are generally small, near-zero, or occasionally positive, suggesting that BTC offers limited potential as a hedge against NASDAQ exposure. For BTC to act as a valid hedging instrument, one would expect negative conditional return correlations coupled with lower-tail dependencies – indicated by the Clayton copula state. Indeed, the estimation results for September 2019 provide some supporting evidence for this, with mildly negative correlations and state probabilities for the Clayton copula, although these probabilities remain moderate (below 40%). A closer look at the correlation patterns and state probabilities reveals a distinct period of high positive return correlation from March to May 2020. During this period, the model assigns relatively low probabilities to the independence state, suggesting that the conclusions drawn from a standard Gaussian BEKK model may overlook important higher-order dependencies. Notably, throughout early 2020, the highest state probabilities are predominantly attributed to states of upper-tail dependence as represented by the Gumbel copula. This suggests that BTC could not be considered an effective hedge against volatility in technology stocks during the onset of the COVID-19 pandemic. Instead, the rise in conditional correlations during this period coincides with increasing probabilities assigned to the Clayton copula state, raising the probability of large (in absolute value), joint negative shocks hitting BTC and NASDAQ. Therefore, BTC did not decouple from broader market trends and should be considered as traditional risky asset, which can be expected to amplify losses rather than providing diversification benefits during crises.

Evidently, the MS-C-MGARCH model proves to be highly responsive to shifts in the dependence structure. While BTC and NASDAQ displayed minimal correlation and were predominantly subject to upper-tail dependence prior to the COVID crisis, the peak of the pandemic saw a significant positive correlation accompanied by a shift towards lower-tail dependence. As the virus spread to central Europe in March 2020, both assets experienced a sharp downturn. On 'Black Thursday', when several Asia-Pacific and European stock markets plummeted, the NASDAQ Composite fell by 5%, and BTC lost nearly half of its value. According to our MS-C-MGARCH model, the state probability of Clapton copula peaks on this occasion. Hence, the joint slump of asset valuation is the result of almost complete positive conditional correlation of returns and a unit probability assigned to a state of lower-tail dependence.

In summary, while the conditional correlation patterns from a standard MGARCH model raise doubts about BTC's effectiveness as a hedge against NASDAQ exposure, the three-state MS-C-MGARCH model offers even stronger evidence to support this skepticism. Specifically, in both the lead-up to and during the pandemic crisis, the model reveals positive correlations alongside shifts to lower-tail dependence states. This suggests that during extreme market downturns, the returns of both assets become increasingly intertwined, exposing them to large, simultaneous negative shocks and thus amplifying the risk of severe, joint price

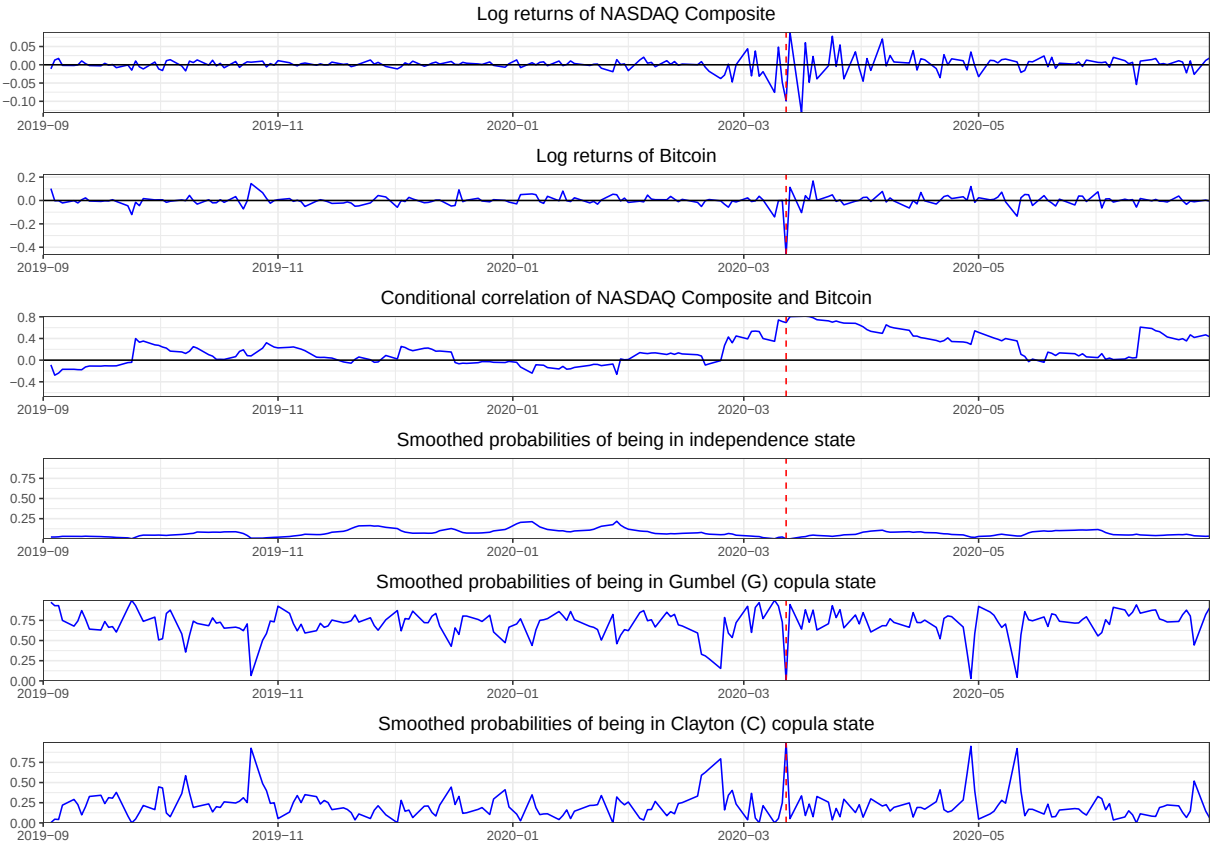


Figure 2: Estimated conditional correlations and smoothed probabilities amids the Covid crisis (time spans from September 1st, 2019 to June 30th, 2020). The red, dashed lines mark the 12th of March 2020 ('Black Thursday').

declines.

5 Conclusion

This paper introduced a new Markov-switching copula-based MGARCH model, which extends the C-MGARCH framework of Lee and Long (2009) by incorporating regime changes to capture potential shifts in the (asymmetric) tail dependence structures among shocks to speculative return series. Specifically, we find that separating correlation from dependence, combined with a parsimonious selection of asymmetric yet complementary higher-order dependence states, allows for a more nuanced analysis of joint return distributions. We propose a two-step QML estimation procedure that is both computationally tractable and consistent, with asymptotic normality under standard assumptions for BEKK and HMM models. In addition to the asymptotic properties, our large-scale simulation study demonstrates that both the transition probabilities and copula shape parameters can be suitably recovered by means of the suggested two-step estimator. Our empirical analysis of NASDAQ and Bitcoin returns offers compelling evidence of higher-order dependence

patterns and their evolution through Markov switching, particularly in the wake of the global pandemic in early 2020. Unlike the Gaussian MGARCH model, which cannot account for tail dependence, and the C-MGARCH model of Lee and Long (2009), which averages out structural changes, our MS-C-MGARCH model captures the significant deviations from the Gaussian benchmark. The results suggest that, rather than providing diversification or serving as a hedge, Bitcoin behaves like a typical risky asset, exacerbating losses during periods of market crisis.

The proposed two-step estimation procedure is well-suited for application to financial time series of higher dimensions, which may enable the detection of shifting dependence structures and improve predictive performance for risk assessment in portfolio management (see Fülle and Herwartz, 2024, for predictive analysis in a three-dimensional context). Future research could explore the possibility of extending the model to investigate Markov-switching in marginal distributions and their parameters, particularly in distinguishing between periods characterized by left- and right-skewed returns.

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References

Ang, A. and J. Chen (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics 63*(3), 443–494.

Ardia, D., K. Bluteau, K. Boudt, and L. Catania (2018). Forecasting risk with Markov-switching GARCH models: A large-scale performance study. *International Journal of Forecasting 34*(4), 733–747.

Azzalini, A. (1985). A class of distributions which includes the normal ones. *Scandinavian Journal of Statistics 12*(2), 171–178.

Bai, X., J. R. Russell, and G. C. Tiao (2003). Kurtosis of GARCH and stochastic volatility models with non-normal innovations. *Journal of Econometrics 114*(2), 349–360.

Bauwens, L., S. Laurent, and J. V. K. Rombouts (2006). Multivariate GARCH models: A survey. *Journal of Applied Econometrics 21*(1), 79–109.

Bauwens, L. and M. Lubrano (1998). Bayesian inference on GARCH models using the Gibbs sampler. *The Econometrics Journal 1*(1), C23–C46.

Bauwens, L., A. Preminger, and J. V. K. Rombouts (2010). Theory and inference for a Markov switching GARCH model. *The Econometrics Journal 13*(2), 218–244.

Berndt, E. R., B. Hall, R. Hall, and J. Hausman (1974). Estimation and inference in nonlinear structural models. In *Annals of Economic and Social Measurement, Volume 3, Number 4*, pp. 653–665. National Bureau of Economic Research, Inc.

Bickel, P. J., Y. Ritov, and T. Rydén (1998). Asymptotic normality of the maximum-likelihood estimator for general hidden Markov models. *The Annals of Statistics 26*(4), 1614 – 1635.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics 31*(3), 307–327.

Bougerol, P. and N. Picard (1992). Strict Stationarity of Generalized Autoregressive Processes. *The Annals of Probability 20*(4), 1714 – 1730.

Boussama, F., F. Fuchs, and R. Stelzer (2011). Stationarity and geometric ergodicity of bekk multivariate garch models. *Stochastic Processes and their Applications 121*(10), 2331–2360.

Broyden, C. G. (1970). The convergence of a class of double-rank minimization algorithms 1. General considerations. *IMA Journal of Applied Mathematics 6*(1), 76–90.

Burda, M. (2015). Constrained Hamiltonian Monte Carlo in BEKK GARCH with targeting. *Journal of Time Series Econometrics 7*(1), 95–113.

Carnero, M. A., D. Peña, and E. Ruiz (2004). Persistence and kurtosis in GARCH and stochastic volatility models. *Journal of Financial Econometrics 2*(2), 319–342.

Chollete, L., A. Heinen, and A. Valdesogo (2009). Modeling international financial returns with a multivariate regime-switching copula. *Journal of Financial Econometrics 7*(4), 437–480.

Christoffersen, P., V. Errunza, K. Jacobs, and H. Langlois (2012). Is the potential for international diversification disappearing? A dynamic copula approach. *Review of Financial Studies 25*(12), 3711–3751.

Comte, F. and O. Lieberman (2003). Asymptotic theory for multivariate GARCH processes. *Journal of Multivariate Analysis 84*(1), 61–84.

Ding, Z. and R. F. Engle (2001). Large scale conditional covariance matrix modeling, estimation and testing. Technical report, NYU working paper No. Fin-01-029.

Drost, F. C. and B. J. Werker (1996). Closing the GARCH gap: Continuous time GARCH modeling. *Journal of Econometrics 74*(1), 31–57.

Engle, R. F. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business and Economic Statistics 20*(3), 339–350.

Engle, R. F. and K. F. Kroner (1995). Multivariate simultaneous generalized ARCH. *Econometric Theory 11*(1), 122–150.

Erb, C. B., C. R. Harvey, and T. E. Viskanta (1994). Forecasting international equity correlations. *Financial Analysts Journal 50*(6), 32–45.

Fengler, M. R., H. Herwartz, and C. Werner (2012). A dynamic copula approach to recovering the index implied volatility skew. *Journal of Financial Econometrics 10*(3), 457–493.

Francq, C. and J.-M. Zakoian (2019). *GARCH models*. John Wiley Sons, Ltd.

Fülle, M. J. and H. Herwartz (2022). Is gold always a safe-haven? Evidence from a novel markov-switching multivariate GARCH model with copula-distributed innovations. Available at SSRN: https://ssrn.com/abstract=3947734.

Fülle, M. J. and H. Herwartz (2024). Predicting tail risks by a markov switching mgarch model with varying copula regimes. *Journal of Forecasting 43*(6), 2163–2186.

Fülle, M. J., A. Lange, C. M. Hafner, and H. Herwartz (2022). BEKKs: An R package for estimation of conditional volatility of multivariate time series.

Fülle, M. J., A. Lange, C. M. Hafner, and H. Herwartz (2024). BEKKs: An R package for estimation of conditional volatility of multivariate time series. *Journal of Statistical Software*.

Garcia, R. and G. Tsafack (2011). Dependence structure and extreme comovements in international equity and bond markets. *Journal of Banking and Finance 35*(8), 1954–1970.

Gray, S. F. (1996). Modeling the conditional distribution of interest rates as a regime-switching process. *Journal of Financial Economics 42*(1), 27–62.

Hafner, C. M. and H. Herwartz (2008). Analytical quasi maximum likelihood inference in multivariate volatility models. *Metrika 67*(2), 219–239.

Hafner, C. M. and A. Preminger (2009). On asymptotic theory for multivariate GARCH models. *Journal of Multivariate Analysis 100*(9), 2044–2054.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica 57*(2), 357.

Hamilton, J. D. (2016). *Regime Switching Models*, pp. 1–7. London: Palgrave Macmillan UK.

Harvey, C. R. and A. Siddique (1999). Autoregressive conditional skewness. *Journal of Financial and Quantitative Analysis 34*(4), 465–487.

Herwartz, H. and F. H. Raters (2015). Copula-MGARCH with continuous covariance decomposition. *Economics Letters 133*(C), 73–76.

Hsu, C. C., C. P. Tseng, and Y. H. Wang (2008). Dynamic hedging with futures: A copula-based GARCH model. *Journal of Futures Markets 28*(11), 1095–1116.

Joe, H. (1996). Families of m-variate distributions with given margins and $m(m-1)/2$ bivariate dependence parameters. In *Institute of Mathematical Statistics Lecture Notes - Monograph Series*, pp. 120–141. Institute of Mathematical Statistics.

Jondeau, E. and M. Rockinger (2006). The copula-GARCH model of conditional dependencies: An international stock market application. *Journal of International Money and Finance 25*(5), 827–853.

Jones, M. C. and M. Faddy (2003). A skew extension of the t-distribution, with applications. *Journal of the Royal Statistical Society: Series B (Statistical Methodology) 65*(1), 159–174.

Klaassen, F. (2002). Improving GARCH volatility forecasts with regime-switching GARCH. *Empirical Economics 27*(2), 363–394.

Kreuzer, A. and C. Czado (2021). Bayesian inference for a single factor copula stochastic volatility model using Hamiltonian Monte Carlo. *Econometrics and Statistics 19*, 130–150.

Lee, T.-H. and X. Long (2009). Copula-based multivariate GARCH model with uncorrelated dependent errors. *Journal of Econometrics 150*(2), 207–218.

Leroux, B. G. (1992). Maximum-likelihood estimation for hidden Markov models. *Stochastic Processes and their Applications 40*(1), 127–143.

Liu, B.-Y., Q. Ji, and Y. Fan (2017). A new time-varying optimal copula model identifying the dependence across markets. *Quantitative Finance 17*(3), 437–453.

Loaiza-Maya, R. and M. S. Smith (2020). Real-time macroeconomic forecasting with a heteroscedastic inversion copula. *Journal of Business & Economic Statistics 38*(2), 470–486.

Longin, F. and B. Solnik (2001). Extreme correlation of international equity markets. *Journal of Finance 56*(2), 649–676.

Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Berlin Heidelberg: Springer.

Magnus, J. R. and H. Neudecker (1999). *Matrix Differential Calculus with Applications in Statistics and Econometrics* (Second ed.). John Wiley.

Meitz, M. and P. Saikkonen (2008). Ergodicity, mixing, and existence of moments of a class of markov models with applications to garch and acd models. *Econometric Theory 24*(5), 1291–1320.

Meyn, S. and R. Tweedie (1993). *Markov Chains and Stochastic Stability*. London: Springer-Verlag.

Nelsen, R. B. (2006). *An Introduction to Copulas*. New York: Springer.

Newey, W. K. and D. McFadden (1994). Chapter 36 Large sample estimation and hypothesis testing. In *Handbook of Econometrics*, pp. 2111–2245. Elsevier.

Patton, A. J. (2004). On the out-of-sample importance of skewness and asymmetric dependence for asset allocation. *Journal of Financial Econometrics 2*(1), 130–168.

Patton, A. J. (2012). A review of copula models for economic time series. *Journal of Multivariate Analysis 110*, 4–18.

Peiró, A. (1999). Skewness in financial returns. *Journal of Banking and Finance 23*(6), 847–862.

Singleton, J. C. and J. Wingender (1986). Skewness persistence in common stock returns. *Journal of Financial and Quantitative Analysis 21*(3), 335–341.

Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. *Publications de Institut Statistique de Universite de Paris 8*, 229–231.

Straumann, D. and T. Mikosch (2006). Quasi-maximum-likelihood estimation in conditionally heteroscedastic time series: A stochastic recurrence equations approach. *The Annals of Statistics 34*(5), 2449 – 2495.

Wang, K., Y. H. Chen, and S. W. Huang (2011). The dynamic dependence between the chinese market and other international stock markets: A time-varying copula approach. *International Review of Economics and Finance 20*(4), 654–664.

Appendix - Proofs of theorems

Proof of Theorem 1: We start by showing that the sequence $\{(r_t, H_t, s_t)\}$ is ergodic and strictly stationary. Denote $h_t := \text{vech}(H_t)$, using the relations $\text{vec}(A) = D_N \text{vech}(A)$, $\text{vech}(A) = D_N^+ \text{vec}(A)$ and $\text{vec}(ABC) = (C' \otimes A)\text{vec}(B)$, the BEKK model in (3) can be written as

$$h_t = Q^* + A^* \text{vech}\left(e_{t-1}e_{t-1}'\right) + G^* h_{t-1}, \tag{14}$$

where $Q^* = D_N^+ \text{vec}(QQ') = D_N^+(Q \otimes Q)\text{vec}(I_N)$, $A^* = D_N^+(A \otimes A)'D_N$ and $G^* = D_N^+(G \otimes G)'D_N$. Thus, the drifting term in the stochastic recurrent equation (11) is

$$\begin{aligned}
\boldsymbol{A}_{t-1} = \frac{\partial h_t}{\partial h_{t-1}} &= A^* D_N^+ \frac{\partial \text{vec}\left(H_{t-1}^{1/2}\eta_{t-1}\eta_{t-1}'H_{t-1}^{1/2'}\right)}{\partial \text{vec}(H_{t-1})} D_N + G^* \\
&= 2A^* D_N^+ N_N (H_t^{1/2}\eta_{t-1} \otimes I_N)(\eta_{t-1}' \otimes I_N) \frac{\partial \text{vec}\left(H_{t-1}^{1/2}\right)}{\partial \text{vec}(H_{t-1})} D_N + G^* \\
&= A^* D_N^+ N_N (H_t^{1/2}\eta_{t-1} \otimes I_N)(\eta_{t-1}' \otimes I_N)(N_N(H_t^{1/2} \otimes I_N))^{-1} D_N + G^* \\
&= A^* D_N^+ \Delta_{t-1}(\phi, \psi) D_N + G^*,
\end{aligned}$$

where N_N is a symmetric idempotent matrix such that $N_N = \frac{1}{2}(I_{N^2} + K_{NN})$ with K_{NN} being the commutation matrix. In addition, $\{s_t\}$ is ergodic and stationary by Assumption A5, which implies by Lemma 1 in Leroux (1992) that $\{\eta_t\}$ is an ergodic strictly stationary martingale difference sequence with marginal distribution given by a finite mixture from a parametric family of copulas. The marginal density is given by $f_\eta = \sum_{k=1}^K \pi(k) f_{\eta_t(k)}$, where $f_{\eta_t(k)}$ is a convolution of at most K absolutely continuous densities and f_η is continuous. By Assumption A3, Minkowski and Cauchy-Schwarz inequalities and since $\pi(k) \in (0,1)$ for all k, we have

$$\mathbb{E}\|\eta_t\|^2 \leq \sum_{k=1}^K \mathbb{E}\|w_k \eta_t(k)\|^2 \leq \sum_{k=1}^K \pi(k)\mathbb{E}\|\eta_t(k)\|^2 \leq \sup_{k\in\{1,\ldots,K\}} \pi(k) \sum_{k=1}^K M_2 \sqrt{\rho(\Sigma(k)^{-1})} \leq K M_1^{1/2} M_2 < \infty.$$

Therefore, the model has a globally attracting state and is forward accessible, and thus by Theorem 7.2.5 and 7.2.6 of Meyn and Tweedie (1993) irreducible. Since $\gamma \leq \mathbb{E}[\log \|\boldsymbol{A}_0\|]$ with equality in case $N = 1$, Assumption A6 implies that the top Lyapunov exponent is strictly negative on parameter space $\mathcal{K}$. Thus, $\{H_t\}$ is geometric ergodic and has a nonanticipative strictly stationary solution, which extends to return sequence $\{r_t\}$ by Proposition 1 of Meitz and Saikkonen (2008).

We next show that the so-called invertibility condition is satisfied such that initial values are asymptoti-

cally irrelevant. By triangle inequality,

$$\sup_{\phi\in\Phi}|\tilde{\mathcal{L}}_T(\phi)-\mathcal{L}_T(\phi)| \le \frac{1}{T}\sum_{t=1}^{T}\sup_{\phi\in\Phi}\left|r_t'\left(H_t^{-1}(\phi)-\tilde{H}_t^{-1}(\phi)\right)r_t\right| + \frac{1}{T}\sum_{t=1}^{T}\sup_{\phi\in\Phi}\left|\log|H_t(\phi)|-\log|\tilde{H}_t(\phi)|\right|$$
$$:= (I)+(II),$$

where

$$\begin{aligned}
(I) &= \frac{1}{T}\sum_{t=1}^{T}\sup_{\phi\in\Phi}\left|r_t'\tilde{H}_t^{-1}(\phi)\left(H_t(\phi)-\tilde{H}_t(\phi)\right)H_t^{-1}(\phi)r_t\right| \\
&= \frac{1}{T}\sum_{t=1}^{T}\sup_{\phi\in\Phi}\left|\mathrm{tr}\left(\tilde{H}_t^{-1}(\phi)\left(H_t(\phi)-\tilde{H}_t(\phi)\right)H_t^{-1}(\phi)r_tr_t'\right)\right| \\
&\le \frac{1}{T}\sum_{t=1}^{T}N\sup_{\phi\in\Phi}\|\tilde{H}_t^{-1}(\phi)\left(H_t(\phi)-\tilde{H}_t(\phi)\right)H_t^{-1}(\phi)\|\|r_tr_t'\| \\
&\le \frac{1}{T}\sum_{t=1}^{T}N\sup_{\phi\in\Phi}\|H_t^{-1}(\phi)\|\|H_t(\phi)-\tilde{H}_t(\phi)\|\|\tilde{H}_t^{-1}(\phi)\|\|r_tr_t'\| \\
&\le T^{-1}NM_hM_{\tilde{h}}M_{h'}\sum_{t=1}^{T}\varrho^t\|r_tr_t'\|,
\end{aligned}$$

where the first inequality holds since for any $A\in\mathbb{R}^{m\times n}$, $B\in\mathbb{R}^{n\times m}$, $|\mathrm{tr}\,(AB)| \le \left(\sum_{ij}a_{ij}^2\right)^{1/2}\left(\sum_{ij}b_{ij}^2\right)^{1/2} = (\mathrm{tr}\,(A'A))^{1/2}(\mathrm{tr}\,(B'B))^{1/2} \le (mn)^{1/2}\|A\|\|B\|$ due to Cauchy-Schwarz inequality, and the last inequality follows from Lemma 5.1 $\varrho\in[0,1)$. Moreover, for some $0<s<1$,

$$T^{-1}\mathbb{E}\left|\sum_{t=1}^{T}\varrho^t\|r_tr_t'\|\right|^s \le T^{-1}\sum_{t=1}^{T}\varrho^{st}\mathbb{E}\|r_t\|^{2s} \le T^{-1}K^2M_1M_2^2\sum_{t=1}^{T}\varrho^{st}\to 0,\ \text{as } T\to\infty.$$

For the second part, since $|A|\le\|A\|^N$ for $A\in R^{N\times N}$ and by Minkowski inequality and Lemma 5.1,

$$\begin{aligned}
\sup_{\phi\in\Phi}\left[\log|H_t(\phi)|-\log|\tilde{H}_t(\phi)|\right] &= \sup_{\phi\in\Phi}\log\left|H_t(\phi)\tilde{H}_t^{-1}(\phi)\right| = \log\left|I_N+(H_t(\phi)-\tilde{H}_t(\phi))\tilde{H}_t^{-1}(\phi)\right| \\
&\le \sup_{\phi\in\Phi}N\log\|I_N+(H_t(\phi)-\tilde{H}_t(\phi))\tilde{H}_t^{-1}(\phi)\| \\
&\le \sup_{\phi\in\Phi}N\log\left(\|I_N\|+\|H_t(\phi)-\tilde{H}_t(\phi)\|\|\tilde{H}_t^{-1}(\phi)\|\right) \\
&\le \sup_{\phi\in\Phi}N\|H_t(\phi)-\tilde{H}_t(\phi)\|\|\tilde{H}_t^{-1}(\phi)\| \\
&\le NM_{h'}M_{\tilde{h}}\varrho^t,
\end{aligned}$$

since $\log x \leq x - 1$, and similarly

$$\sup_{\phi\in\Phi}\left[\log|H_t(\phi)| - \log|\tilde{H}_t(\phi)|\right] \leq \sup_{\phi\in\Phi} N\|H_t(\phi) - \tilde{H}_t(\phi)\|\|H_t^{-1}(\phi) \leq NM_{h'}M_h\varrho^t.$$

Thus,

$$(II) \leq T^{-1}NM_{h'}\max\{M_h, M_{\tilde{h}}\}\sum_{t=1}^{T}\varrho^t \to 0, \text{ as } T\to\infty,$$

and we obtain $\sup_{\phi\in\Phi}|\tilde{\mathcal{L}}_T(\phi) - \mathcal{L}_T(\phi)| \to 0$, as $T\to\infty$.

Finally, since both l_t and $\tilde{l}_t$ are continuous and thus measurable functions of the ergodic strictly stationary sequence $\{\eta_u : u \leq t\}$, sequences $\{l_t\}$, $\{\tilde{l}_t\}$ are also ergodic strictly stationary. Since $\lim_{T\to\infty}\sup_{\phi\in\Phi}|\tilde{\mathcal{L}}_T(\phi) - \mathcal{L}_T(\phi)| \to 0$, a.s., i.e., they have the same limit, applying a uniform ergodic theorem entails that $\sup_{\phi\in\Phi}|\mathcal{L}_T(\phi) - \mathbb{E}[-\tilde{l}_t(\phi)]| = 0$, a.s., as $T\to\infty$. By Lemma 5.2, the population objective function $\mathbb{E}[\tilde{l}_t(\phi)]$ has a clear and unique minimizer at the true value ϕ_0. This proves the consistency of the first-step QML estimator $\hat{\phi}_{T,QML}$.

To prove the consistency of the second-step CML estimator, we define $\eta_t := \eta_t(\phi_0) = H_t^{-1/2}(\phi_0)r_t$ for each t and write

$$\mathcal{G}_T(\psi) = \mathcal{G}_T(\psi,\phi_0) = \frac{1}{T}\log g_T(\eta_1,\ldots,\eta_T;\psi) = \frac{1}{T}\sum_{t=1}^{T}\log g_t(\eta_t|\eta_{t-1},\ldots,\eta_1;\psi) = \frac{1}{T}\sum_{t=1}^{T}\log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1};\psi),$$

where $g_T : \mathbb{R}^N\times\Psi\mapsto[0,\infty)$ and $g_t : \mathbb{R}\times\Psi\mapsto[0,\infty)$ are marginal and conditional density functions with respect to ν^N and ν, receptively, and $\tilde{\Omega}_t^u = \boldsymbol{\sigma}(\{\eta_t,\eta_{t-1},\ldots,\eta_{t-u}\})$. Denote $\tilde{\Omega}_t^\infty = \lim_{u\to\infty}\tilde{\Omega}_t^u$ the sigma-filed generated by the infinite past, given the uniform integrability provided by Assumption A3, we have

$$\sup_{\psi\in\Psi}\left|\mathcal{G}_T(\psi) - \mathbb{E}[\log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1};\psi)]\right| \to 0, \ a.s.,$$

as $T\to\infty$. Since for all $t = 1,\ldots,T$, we have by Assumption A3

$$-\infty < \inf_{k\in\{1,\ldots,K\}}\left(\frac{1}{2}\log|\Sigma(k;\psi)| + \log f_k\left(\Sigma(k;\psi)^{1/2}\eta_t;\psi\right)\right) \leq \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1};\psi) \leq$$
$$\sup_{k\in\{1,\ldots,K\}}\left(\frac{1}{2}\log|\Sigma(k;\psi)| + \log f_k\left(\Sigma(k;\psi)^{1/2}\eta_t;\psi\right)\right) < \infty,$$

thus $\mathbb{E}[\log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1};\psi)]$ is well-defined on Ψ. Due to stationarity as implied by Assumption A5 and discussed above $\mathbb{E}[\log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1};\psi)] = \mathbb{E}[\log g_1(\eta_1|\tilde{\Omega}_0^{t-1};\psi)] = \mathbb{E}[\log g_1(\eta_1|\eta_0,\ldots,\eta_{-t+2};\psi)]$, where

$$g_1(\eta_1|\tilde{\Omega}_0^{t-1};\psi) = \sum_{k=1}^{K}\mathbb{P}(s_1 = k|\tilde{\Omega}_0^{t-1})|\Sigma(k)^{1/2}|f_k\left(\Sigma(k;\psi)^{1/2}\eta_1;\psi\right).$$

By a classical marginale convergence result, $\lim_{t\to\infty} g_1(\eta_1|\tilde{\Omega}_0^{t-1};\psi) = g(\eta_1|\tilde{\Omega}_0^{\infty};\psi)$, and thus as $T\to\infty$

$$\frac{1}{T}\sum_{t=1}^{T}\log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1};\psi) \to \mathbb{E}[g(\eta_1|\tilde{\Omega}_0^{\infty};\psi)],\ a.s.$$

Now we show that choice of initial distribution $\pi^*(k)$ of the chain is irrelevant asymptotically. Let

$$\tilde{g}_T^{(k)}(\eta_1,\ldots,\eta_T;\psi) = |\Sigma(k)^{1/2}| f_k\left(\Sigma(k;\psi)^{1/2}\eta_1;\psi\right)\sum_{s_2=1}^{K}\cdots\sum_{s_T=1}^{K} p_{s_2k}|\Sigma(s_2;\psi)^{1/2}| f_{s_2}\left(\Sigma(s_2;\psi)^{1/2}\eta_2;\psi\right)$$
$$\prod_{t=3}^{T} p_{s_t s_{t-1}}|\Sigma(s_t;\psi)^{1/2}| f_{s_t}\left(\Sigma(s_t;\psi)^{1/2}\eta_t;\psi\right),$$

be the joint density for a fixed initial state $s_1 = k$. Note that

$$g_T(\eta_1,\ldots,\eta_T;\psi) = \sum_{k=1}^{K}\pi^*(k)\tilde{g}_T^{(k)}(\eta_1,\ldots,\eta_T;\psi) \le \bar{g}_T(\eta_1,\ldots,\eta_T;\psi)\underbrace{\sum_{k=1}^{K}\pi^*(k)}_{=1},$$

where $\bar{g}_T(\eta_1,\ldots,\eta_T;\psi) = \sup_{k\in\{1,\ldots,K\}}\tilde{g}_T^{(k)}(\eta_1,\ldots,\eta_T;\psi)$. Moreover, let $\underline{\pi}^* = \inf_{k\in\{1,\ldots,K\}}\pi^*(k) > 0$, then $g_T(\eta_1,\ldots,\eta_T;\psi) \ge \underline{\pi}^*\bar{g}_T(\eta_1,\ldots,\eta_T;\psi)$. Therefore, for all ψ on Ψ,

$$\underline{\pi}^*\bar{g}_T(\eta_1,\ldots,\eta_T;\psi) \le g_T(\eta_1,\ldots,\eta_T;\psi) \le \bar{g}_T(\eta_1,\ldots,\eta_T;\psi)$$

and thus $\log\underline{\pi}^* \le \log g_T(\eta_1,\ldots,\eta_T;\psi) - \log\bar{g}_T(\eta_1,\ldots,\eta_T;\psi) \le 0$. Note that the log-likelihood function $\log\bar{g}_T(\eta_1,\ldots,\eta_T;\psi)$ does not depend on the initial distribution. Moreover, $\log\bar{g}_T(\eta_1,\ldots,\eta_T;\psi)$ is integrable by Assumption A3 and the sequence is stationary sequence and satisfies the so-called sub-additivity (see Lemma 3 in Leroux 1992), thus $T^{-1}\log\bar{g}_T(\eta_1,\ldots,\eta_T;\psi)$ converges to the same limit as $T^{-1}\log g_T(\eta_1,\ldots,\eta_T;\psi) = \mathcal{G}_T(\psi)$ by Theorem 2 of Leroux (1992). Furthermore, under Assumption A2, A5 and A3, for any $\delta > 0$, there exists an open ball $B(\psi_0,\delta)$, s.t.,

$$\sup_{\phi\in B(\psi_0,\delta)^c\cap\Psi}\mathbb{E}[g(\eta_1|\tilde{\Omega}_0^{\infty};\psi)] < \mathbb{E}[g(\eta_1|\tilde{\Omega}_0^{\infty};\psi_0)],$$

by Lemma 6 of of Leroux (1992). Therefore, as $T\to\infty$,

$$\hat{\psi}_{T,CML}(\phi_0) = \underset{\psi\in\Psi}{\operatorname{argmax}}\,\mathcal{G}_T(\psi,\phi_0) \to \psi_0,\ a.s.$$

Since the first-step estimator is strongly consistent, as $T\to\infty$, $\hat{\phi}_{T,QML}\to\phi_0$, a.s., by a continuous mapping theorem, $\eta_t(\hat{\phi}_{T,QML})\to\eta_t$, a.s., such that the objective function $\mathcal{G}_T(\psi,\hat{\phi}_{T,QML})$ coincides with $\mathcal{G}_T(\psi)$ as T

grows to infinity and $\hat{\psi}_{T,CML} = \hat{\psi}_{T,CML}(\hat{\phi}_{T,QML})$ converges to the same limit. This completes the proof. □

Proof of Theorem 2: We start by proving the asymptotic normality of the first-step estimator. Conditional on the initial values generated from the stationary distribution, for $i = 1, \ldots, d_\phi$, the corresponding (negative) score is given by

$$\frac{\partial \tilde{l}_t(\phi)}{\partial \phi_i} = \frac{\partial}{\partial \phi_i} r_t' \tilde{H}_t^{-1}(\phi) r_t + \frac{\partial}{\partial \phi_i} \log |H_t(\phi)| = \text{tr}\left(-\tilde{H}_t^{-1}(\phi) r_t r_t' \tilde{H}_t^{-1}(\phi) \frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i} + \tilde{H}_t^{-1}(\phi) \frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i} \right).$$

Thus, evaluating at ϕ_0 and using the elementary relation $\text{vec}\,(A)'\,\text{vec}\,(B) = \text{tr}\,(A'B)$, we have

$$\begin{aligned}
\frac{\partial \tilde{l}_t(\phi_0)}{\partial \phi_i} &= \text{tr}\left(\tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} - \tilde{H}_t^{-1}(\phi_0) \tilde{H}_t^{1/2}(\phi_0) \eta_t \eta_t' \tilde{H}_t^{1/2'}(\phi_0) \tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \right) \\
&= \text{tr}\left(\tilde{H}_t^{-1/2}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \tilde{H}_t^{-1/2'}(\phi_0) - \eta_t \eta_t' \tilde{H}_t^{-1/2}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \tilde{H}_t^{-1/2'}(\phi_0) \right) \\
&= \text{tr}\left((I_N - \eta_t \eta_t') \tilde{H}_t^{-1/2}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \tilde{H}_t^{-1/2'}(\phi_0) \right) \\
&= \frac{\partial \text{vec}\left(\tilde{H}_t(\phi_0) \right)'}{\partial \phi_i} (\tilde{H}_t^{-1/2}(\phi_0) \otimes \tilde{H}_t^{-1/2}(\phi_0))' \text{vec}\,(I_N - \eta_t \eta_t').
\end{aligned}$$

Let $\mathcal{I}_{i,j} = \mathbb{E}\left[\frac{\partial \tilde{l}_t(\phi_0)}{\partial \phi_i} \frac{\partial \tilde{l}_t(\phi_0)}{\partial \phi_j}' | \Omega_{t-1} \right]$, for $i, j \in \{1, \ldots, d_\phi\}$

$$\begin{aligned}
\mathcal{I}_{i,j} = \mathbb{E}\Bigg[&\text{tr}\Bigg(\text{vec}\,(I_N - \eta_t \eta_t')\,\text{vec}\,(I_N - \eta_t \eta_t')' (\tilde{H}_t^{-1/2}(\phi_0) \otimes \tilde{H}_t^{-1/2}(\phi_0)) \frac{\partial \text{vec}\left(\tilde{H}_t(\phi_0) \right)}{\partial \phi_j} \\
&\frac{\partial \text{vec}\left(\tilde{H}_t(\phi_0) \right)'}{\partial \phi_i} (\tilde{H}_t^{-1/2}(\phi_0) \otimes \tilde{H}_t^{-1/2}(\phi_0))' \Bigg) \Bigg] \\
= \text{tr}\Big\{ &\mathbb{E}\left[\text{vec}\,(I_N - \eta_t \eta_t')\,\text{vec}\,(I_N - \eta_t \eta_t')' \right] \\
&\mathbb{E}\left[(\tilde{H}_t^{-1/2}(\phi_0) \otimes \tilde{H}_t^{-1/2}(\phi_0)) \frac{\partial \text{vec}\left(\tilde{H}_t(\phi_0) \right)}{\partial \phi_j} \frac{\partial \text{vec}\left(\tilde{H}_t(\phi_0) \right)'}{\partial \phi_i} (\tilde{H}_t^{-1/2}(\phi_0) \otimes \tilde{H}_t^{-1/2}(\phi_0))' \right] \Big\},
\end{aligned}$$

where the last equality follows from the conditional independence between η_t and H_t (and its derivative) and the conditional information set in is dropped to simply the notation. Denote $M_\eta := \mathbb{E}\left[\text{vec}\,(I_N - \eta_t \eta_t')\,\text{vec}\,(I_N - \eta_t \eta_t')' \right]$. Since by Lemma 5.1 and a similar argument in the proof of Theorem 1,

$$\mathbb{E}\|\eta_t\|^4 = \mathbb{E}\left[\text{tr}\,(\eta_t' \eta_t)^2 \right] = \mathbb{E}\left[\text{tr}\left(r_t' \tilde{H}_t^{-1}(\phi_0) r_t \right)^2 \right] \leq N^2 M_{\tilde{h}}^2 \mathbb{E}\|r_t\|^4 < \infty$$

due to Assumption A8, $M_\eta < \infty$. By Lemma 5.3, $\mathbb{E}\left[\sup_{\phi\in B(\phi_0,\delta)}\left\|\tilde{H}_t^{-1/2}(\phi)\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i}\tilde{H}_t^{-1/2'}(\phi)\right\|^{2s}\right] < \infty$ for some $s > 1$, for any $i \in \{1, \ldots, d_\phi\}$,

$$\mathbb{E}\left\|(\tilde{H}_t^{-1/2}(\phi_0) \otimes \tilde{H}_t^{-1/2}(\phi_0))\frac{\partial \text{vec}\left(\tilde{H}_t(\phi_0)\right)}{\partial \phi_i}\right\|^2 < \infty.$$

Therefore, the information matrix $\mathcal{I} = (\mathcal{I}_{i,j})$ is well-defined. Moreover, by definition, since $\tilde{H}_t(\phi_0)$ is the conditional covariance matrix of r_t, we have $\mathbb{E}\left[\frac{\partial \tilde{l}_t(\phi_0)}{\partial \phi_i}|\Omega_{t-1}\right] = 0$ for all i. Thus, under our assumptions, $\left\{\frac{\partial \tilde{l}_t(\phi_0)}{\partial \phi}\right\}$ is a square integrable stationary ergodic martingale difference sequence. By a corresponding CLT and Cramér-Wold device, as $T \to \infty$,

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T}\frac{\partial \tilde{l}_t(\phi_0)}{\partial \phi} \xrightarrow{d} \mathcal{N}(0, \mathcal{I}).$$

By Lemma 3 in Hafner and Preminger (2009) and Lemma 5.3, by a similar argument in the proof of Theorem 1 it can be shown that as $T \to \infty$,

$$\sup_{\phi\in B(\phi_0,\delta)}\sqrt{T}\left\|\frac{\partial \tilde{\mathcal{L}}_T(\phi)}{\partial \phi} - \frac{\partial \mathcal{L}_T(\phi)}{\partial \phi}\right\| \to 0, \ a.s.$$

For any $i, j \in \{1, \ldots, d_\phi\}$, the Hessian is given by

$$\begin{aligned}
\frac{\partial^2 \tilde{l}_t(\phi)}{\partial \phi_i \partial \phi_j} &= \frac{\partial}{\partial \phi_j}\text{tr}\left(-\tilde{H}_t^{-1}(\phi)r_t r_t'\tilde{H}_t^{-1}(\phi)\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i} + \tilde{H}_t^{-1}(\phi)\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i}\right) \\
&= \underbrace{r_t'\tilde{H}_t^{-1}(\phi)\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i}\tilde{H}_t^{-1}(\phi)\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_j}\tilde{H}_t^{-1}(\phi)r_t - \text{tr}\left(\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i}\tilde{H}_t^{-1}(\phi)\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_j}\tilde{H}_t^{-1}(\phi)\right)}_{:=\mathcal{J}_{t,I}^*(\phi)} \\
&\quad + \underbrace{r_t'\tilde{H}_t^{-1}(\phi)\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_j}\tilde{H}_t^{-1}(\phi)\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i}\tilde{H}_t^{-1}(\phi)r_t}_{:=\mathcal{J}_{t,II}^*(\phi)} \\
&\quad + \underbrace{\text{tr}\left(\tilde{H}_t^{-1}(\phi)\frac{\partial^2 \tilde{H}_t(\phi)}{\partial \phi_i \partial \phi_j}\right) - r_t'\tilde{H}_t^{-1}(\phi)\frac{\partial^2 \tilde{H}_t(\phi)}{\partial \phi_i \partial \phi_j}\tilde{H}_t^{-1}(\phi)r_t}_{:=\mathcal{J}_{t,III}^*(\phi)}.
\end{aligned}$$

Note that at the true parameter value, $\mathbb{E}[\tilde{H}_t^{-1}(\phi_0) r_t r_t'] = I_N$, thus for $\mathcal{J}^*_{t,I}(\phi_0)$ and $\mathcal{J}^*_{t,III}(\phi_0)$,

$$\begin{aligned}
\mathbb{E}\left[r_t' \tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_j} \tilde{H}_t^{-1}(\phi_0) r_t \right] &= \mathbb{E}\left[\operatorname{tr}\left(\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i} \tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_j} \tilde{H}_t^{-1}(\phi_0) r_t r_t' \tilde{H}_t^{-1}(\phi_0) \right) \right] \\
&= \mathbb{E}\left[\operatorname{tr}\left(\frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i} \tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_j} \tilde{H}_t^{-1}(\phi_0) \right) \right] \\
\mathbb{E}\left[r_t' \tilde{H}_t^{-1}(\phi_0) \frac{\partial^2 \tilde{H}_t(\phi_0)}{\partial \phi_i \partial \phi_j} \tilde{H}_t^{-1}(\phi_0) r_t \right] &= \mathbb{E}\left[\operatorname{tr}\left(\tilde{H}_t^{-1}(\phi_0) \frac{\partial^2 \tilde{H}_t(\phi_0)}{\partial \phi_i \partial \phi_j} \tilde{H}_t^{-1}(\phi_0) r_t r_t' \right) \right] \\
&= \mathbb{E}\left[\operatorname{tr}\left(\tilde{H}_t^{-1}(\phi_0) \frac{\partial^2 \tilde{H}_t(\phi_0)}{\partial \phi_i \partial \phi_j} \right) \right].
\end{aligned}$$

By Lemma 5.3, both $\mathbb{E}\left[\sup_{\phi \in B(\phi_0,\delta)} \|\mathcal{J}^*_{t,I}(\phi)\|\right]$ and $\mathbb{E}\left[\sup_{\phi \in B(\phi_0,\delta)} \|\mathcal{J}^*_{t,III}(\phi)\|\right]$ bounded, therefore

$$\sup_{\phi \in B(\phi_0,\delta)} \left| \frac{1}{T} \sum_{t=1}^{T} \left(\mathcal{J}^*_{t,I}(\phi) + \mathcal{J}^*_{t,III}(\phi) \right) \right| \to 0, \ a.s.,$$

by ergodic theorem. Moreover,

$$\begin{aligned}
\mathbb{E}[\mathcal{J}^*_{t,II}(\phi_0)] &= \operatorname{tr}\left(\mathbb{E}\left[\frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_j} \tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \tilde{H}_t^{-1}(\phi_0) r_t r_t' \tilde{H}_t^{-1}(\phi_0) \right] \right) \\
&= \mathbb{E}\left[\operatorname{tr}\left(\frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_j} \tilde{H}_t^{-1}(\phi_0) \right) \right] \\
&= \mathbb{E}\left[\operatorname{vec}\left(\frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \right)' \operatorname{vec}\left(\tilde{H}_t^{-1}(\phi_0) \frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_j} \tilde{H}_t^{-1}(\phi_0) \right) \right] \\
&= \mathbb{E}\left[\operatorname{vec}\left(\frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_i} \right)' (\tilde{H}_t^{-1}(\phi_0) \otimes \tilde{H}_t^{-1}(\phi_0)) \operatorname{vec}\left(\frac{\partial \tilde{H}_t(\phi_0)}{\partial \phi_j} \right) \right] \\
&:= \mathcal{J}_{i,j}.
\end{aligned}$$

By Lemma 5.3, the dominating function for $\mathcal{J}^*_{t,II}(\phi)$ is also integrable within a neighborhood about ϕ_0 and thus for all $i, j \in \{1, \ldots, d_\phi\}$, the limit $\mathcal{J}_{i,j}$ is well-defined and by ergodic theorem

$$\sup_{\phi \in B(\phi_0,\delta)} \left| \frac{1}{T} \sum_{t=1}^{T} \frac{\partial^2 \tilde{l}_t(\phi)}{\partial \phi_i \partial \phi_j} - \mathcal{J}_{i,j} \right| \to 0, \ a.s.$$

By Lemma 5.3 and using a similar argument for the score, it can be shown that as $T \to \infty$

$$\sup_{\phi \in B(\phi_0,\delta)} \left\| \frac{\partial^2 \tilde{\mathcal{L}}_T(\phi)}{\partial \phi \partial \phi'} - \frac{\partial^2 \mathcal{L}_T(\phi)}{\partial \phi \partial \phi'} \right\| \to 0, \ a.s.$$

By definition of $\hat{\phi}_{T,QML}$, the first order condition $\frac{1}{\sqrt{T}}\sum_{t=1}^{T} l_t(\hat{\phi}_{T,QML})/\partial\phi = 0$ is satisfied with probability approaching one, by a mean value theorem

$$0 = \frac{1}{\sqrt{T}}\sum_{t=1}^{T}\frac{l_t(\phi_0)}{\partial\phi} + \frac{1}{T}\sum_{t=1}^{T}\frac{\partial^2 l_t(\phi^*)}{\partial\phi\partial\phi'}\sqrt{T}(\hat{\phi}_{T,QML} - \phi_0),$$

where ϕ^* is a (component-wise) mean value. By the results above and Theorem 1, $\frac{1}{T}\sum_{t=1}^{T}\frac{\partial^2 l_t(\phi^*)}{\partial\phi\partial\phi'} \to \mathcal{J}$ a.s., where $\mathcal{J} = (\mathcal{J}_{i,j})$ is the $d_\phi \times d_\phi$ Hessian matrix with the $[i,j]$-component given above. Thus, as $T \to \infty$

$$\sqrt{T}(\hat{\phi}_{T,QML} - \phi_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}).$$

To derive the asymptotic distribution of the second-step estimator, we use the notations introduced in the proof of Theorem 1 and additionally define $\dot{\sigma}_i(k) = \frac{\partial}{\partial\psi_i}\text{vech}\left(\Sigma(k;\psi_0)^{1/2}\right)$, $\dot{\sigma}(k) = (\dot{\sigma}_1(k),\ldots,\dot{\sigma}_{d_\psi}(k))'$, $\alpha(k,l) = \frac{\partial}{\partial\psi}\log p_{lk}(\psi_0)$, $\beta(\eta_t|k) = \frac{\partial}{\partial\psi}\log f_{\eta_t(k)}(\eta_t;\theta_{0,k})$. Notably, for $l,k \in \{1,\ldots,K\}$,

$$\alpha(k,l) = \left(\mathbf{0}'_{1\times\sum_{k=1}^{K} d_{\theta_k}}, 0,\ldots,0,p_{lk}^{-1},0,\ldots,0\right)'$$

and $\beta(\eta_t|k) = (\beta_1(\eta_t|k),\ldots,\beta_{d_\phi}(\eta_t|k))'$ with $i \in \{1,\ldots,d_\phi\}$,

$$\begin{aligned}
\beta_i(\eta_t|k) &= \frac{\partial}{\partial\psi}\log f_{\eta_t(k)}(\eta_t;\theta_{0,k}) = \frac{\partial}{\partial\psi}\text{tr}\left(\log|\Sigma(k;\psi_0)^{1/2}| + \log f_k(\Sigma(k;\psi_0)^{1/2}\eta_t;\theta_{0,k})\right)\\
&= \text{tr}\left(\Sigma(k;\psi_0)^{-1/2}\frac{\partial\Sigma(k;\psi_0)^{1/2}}{\partial\psi_i} + \frac{f_k'(\xi_t(k);\theta_{0,k})'}{f_k(\xi_t(k);\theta_{0,k})}(\eta_t'\otimes I_N)\frac{\partial\text{vec}\left(\Sigma(k;\psi_0)^{1/2}\right)}{\partial\psi_i}\right)\\
&= \text{vec}\left(\Sigma(k;\psi_0)^{-1/2}\right)'\frac{\partial\text{vec}\left(\Sigma(k;\psi_0)^{1/2}\right)}{\partial\psi_i} + \frac{f_k'(\xi_t(k);\theta_{0,k})'}{f_k(\xi_t(k);\theta_{0,k})}(\eta_t'\otimes I_N)\frac{\partial\text{vec}\left(\Sigma(k;\psi_0)^{1/2}\right)}{\partial\psi_i}\\
&= \left[\text{vec}\left(\Sigma(k;\psi_0)^{-1/2}\right)' + \frac{f_k'(\xi_t(k);\theta_{0,k})'}{f_k(\xi_t(k);\theta_{0,k})}(\eta_t'\otimes I_N)\right]D_N\dot{\sigma}_i(k),
\end{aligned}$$

where $f_k'(\xi_t(k);\theta_{0,k}) = \left(\frac{\partial}{\partial\xi_{t,1}(k)}f_k(\xi_t(k);\theta_{0,k}),\ldots,\frac{\partial}{\partial\xi_{t,N}(k)}f_k(\xi_t(k);\theta_{0,k})\right)'$ and we write $\xi_t(k) = \Sigma(k;\psi_0)^{1/2}\eta_t$. Now, define sequence $\{\zeta_t\}_{t=1}^{T}$ such that

$$\zeta_t = \lim_{u\to\infty}\sum_{\tau=-u}^{t-1}\left(\mathbb{E}\left[\beta(\eta_\tau|s_\tau) + \alpha(s_\tau,s_{\tau+1})|\tilde{\Omega}_t^{\infty}\right] - \mathbb{E}\left[\beta(\eta_\tau|s_\tau) + \alpha(s_\tau,s_{\tau+1})|\tilde{\Omega}_{t-1}^{\infty}\right]\right) + \mathbb{E}\left[\beta(\eta_t|s_t)|\tilde{\Omega}_t^{\infty}\right].$$

Bickel et al. (1998) shown that $\{\zeta_t\}_{t=1}^{T}$ is a stationary ergodic martingale difference sequence and by a CLT

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T}\zeta_t \xrightarrow{d} \mathcal{N}(0,\mathcal{V}),$$

where $\mathcal{V} := \mathbb{E}[\zeta_t \zeta_t']$. Recall that the objective function is $\mathcal{G}_T(\psi) = \mathcal{G}_T(\psi, \phi_0) = \frac{1}{T}\sum_{t=1}^{T} \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi)$, under Assumption A9 by Lemma 6 in Bickel et al. (1998), as $T \to \infty$,

$$\left\| \frac{\partial \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi} - \zeta_t \right\| \to 0, \ a.s.$$

exponentially fast and thus $T^{-1/2}\sum_{t=1}^{T} \zeta_t$ has the same limiting distribution as $T^{-1/2}\sum_{t=1}^{T} \frac{\partial \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi}$. Therefore, as $T \to \infty$,

$$\sqrt{T}\frac{\partial \mathcal{G}_T(\psi_0)}{\partial \psi} = \frac{1}{\sqrt{T}}\sum_{t=1}^{T} \frac{\partial \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi} \xrightarrow{d} \mathcal{N}(0, \mathcal{V}).$$

Under Assumption A9 and by Lemma 2 of Bickel et al. (1998),

$$\mathbb{E}\left[\frac{\partial^2 \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi \partial \psi'}\right] = -\mathbb{E}\left[\frac{\partial \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi}\left(\frac{\partial \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi}\right)'\right] = -\mathcal{V}.$$

Given the consistency of first-step estimator shown in Theorem 1, by a mean value theorem, we have with probability approaching one,

$$0 = \frac{1}{\sqrt{T}}\sum_{t=1}^{T} \frac{\partial \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi} + \frac{1}{T}\sum_{t=1}^{T} \frac{\partial^2 \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi^*)}{\partial \psi \partial \psi'}\sqrt{T}(\hat{\psi}_{T,CML}(\phi_0) - \psi_0),$$

where ψ^* is a mean value between $\hat{\psi}_{T,CML}$ and ψ_0 such that $\psi^* \to \psi_0$, a.s., as $T \to \infty$ due to Theorem 1. By Lemma 10 of Bickel et al. (1998) and ergodic theorem, $-\frac{1}{T}\sum_{t=1}^{T} \frac{\partial^2 \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi^*)}{\partial \psi \partial \psi'} \to \mathcal{V}$ a.s., and therefore, as $T \to \infty$,

$$\sqrt{T}(\hat{\psi}_{T,CML}(\phi_0) - \psi_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{V}^{-1}).$$

To derive the joint asymptotic distribution of $\hat{\kappa}_{T,2SE} = \left(\hat{\phi}_{T,QML}', \hat{\psi}_{T,CML}'\right)'$, note that $\left(\hat{\phi}_{T,QML}', \hat{\psi}_{T,CML}'\right)'$ solves the system of equations $\sqrt{T}\left(\frac{\partial \mathcal{L}_T(\phi)}{\partial \phi}', \frac{\partial \mathcal{G}_T(\psi,\phi)}{\partial \psi}'\right)' = \mathbf{0}_{d_\kappa \times 1}$. Denote the first order condition of the population objective function $\mathbb{E}[\boldsymbol{m}(\psi_0, \phi_0)] = \mathbf{0}_{d_\kappa \times 1}$ and define $\mathcal{W} = \mathbb{E}\left[\frac{\partial l_t(\phi_0)}{\partial \phi}\left(\frac{\partial \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi}\right)'\right]$ to be a $d_\phi \times d_\psi$ matrix. Let $\mathcal{H} = \mathbb{E}\left[\frac{\partial^2 \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi \partial \phi'}\right]$ be the $d_\psi \times d_\phi$ expected Jacobian such that $\mathcal{H} = (\mathcal{H}_{i,j})$ with $\mathcal{H}_{i,j} = \mathbb{E}\left[\frac{\partial^2 \log g_t(\eta_t|\tilde{\Omega}_{t-1}^{t-1}; \psi_0)}{\partial \psi_i \partial \phi_j}\right]$, which is well-defined and can be computed using the results above by

computing the Hessian of density $f_k(\xi_t(k);\theta_{0,k})$ and by noting that

$$\begin{aligned}\frac{\partial \xi_t(k)}{\partial \phi_j} &= \Sigma(k;\psi_0)^{1/2}\frac{\partial \eta_t(\phi_0)}{\partial \phi_j} = \Sigma(k;\psi_0)^{1/2}(r_t'\otimes I_N)\frac{\partial \text{vec}\left(H_t^{-1/2}(\phi_0)\right)}{\partial \phi_j}\\ &= -\frac{1}{2}\Sigma(k;\psi_0)^{1/2}(r_t'\otimes I_N)(H_t^{-1/2'}(\phi_0)\otimes H_t^{-1/2}(\phi_0))'(N_N(H_t(\phi_0)\otimes I_N))^{-1}\frac{\partial \text{vec}\left(H_t(\phi_0)\right)}{\partial \phi_j}.\end{aligned}$$

Therefore, the joint asymptotic covariance matrix is (see also Theorem 6.1 in Newey and McFadden, 1994)

$$\begin{aligned}\mathbb{E}\left[T\left(\hat{\kappa}_{2SE}-\kappa_0\right)\left(\hat{\kappa}_{2SE}-\kappa_0\right)'\right] &\to \mathbb{E}\left[\frac{\partial \boldsymbol{m}(\psi_0,\phi_0)}{\partial(\phi',\psi')'}\right]^{-1}\mathbb{E}\left[\boldsymbol{m}(\psi_0,\phi_0)\boldsymbol{m}(\psi_0,\phi_0)'\right]\mathbb{E}\left[\frac{\partial \boldsymbol{m}(\psi_0,\phi_0)}{\partial(\phi',\psi')'}\right]^{-1'}\\ &= \begin{pmatrix}\mathcal{J}^{-1} & \mathbf{0}_{d_\phi\times d_\psi}\\ \mathcal{V}^{-1}\mathcal{H}\mathcal{J}^{-1} & -\mathcal{V}^{-1}\end{pmatrix}\begin{pmatrix}\mathcal{I} & \mathcal{W}\\ \mathcal{W}' & \mathcal{V}\end{pmatrix}\begin{pmatrix}\mathcal{J}^{-1} & \mathcal{J}^{-1}\mathcal{H}'\mathcal{V}^{-1}\\ \mathbf{0}'_{d_\psi\times d_\phi} & -\mathcal{V}^{-1}\end{pmatrix}\\ &= \begin{pmatrix}\mathcal{M}_{1,1} & \mathcal{M}_{1,2}\\ \mathcal{M}_{2,1} & \mathcal{M}_{2,2}\end{pmatrix},\ a.s.,\end{aligned}$$

where $\mathcal{M}_{1,1}=\mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}$, $\mathcal{M}_{1,2}=\mathcal{M}_{1,1}\mathcal{H}'\mathcal{V}^{-1}-\mathcal{J}^{-1}\mathcal{W}\mathcal{V}^{-1}$, $\mathcal{M}_{2,1}=\mathcal{V}^{-1}\mathcal{H}\mathcal{M}_{1,1}-\mathcal{V}^{-1}\mathcal{W}'\mathcal{J}^{-1}$ and $\mathcal{M}_{2,2}=\mathcal{V}^{-1}\mathcal{H}\mathcal{M}_{1,1}\mathcal{H}'\mathcal{V}^{-1}-\mathcal{V}^{-1}\mathcal{H}\mathcal{J}^{-1}\mathcal{W}\mathcal{V}^{-1}-\mathcal{V}^{-1}\mathcal{W}'\mathcal{J}^{-1}\mathcal{H}'\mathcal{V}^{-1}+\mathcal{V}^{-1}$. This completes the proof. □

Lemma 5.1. *There exist* $0<M_h,M_{\tilde{h}},M_{h'}<\infty$, $\varrho\in[0,1)$, *s.t.,*

$$\sup_{\phi\in\Phi}\|H_t^{-1}(\phi)\|\leq M_h,\ \sup_{\phi\in\Phi}\|\tilde{H}_t^{-1}(\phi)\|\leq M_{\tilde{h}},\ \textit{and}\ \ \sup_{\phi\in\Phi}\|\tilde{H}_t(\phi)-H_t(\phi)\|\leq \varrho^t M_{h'},\ a.s.$$

Proof. First, write $H_t(\phi)=S_t(\phi)R_t(\phi)S_t(\phi)$, where $S_t(\phi)$ is a diagonal matrix collecting conditional standard deviations of the returns $S_t(\phi)=\text{diag}\left(\sigma_{r1,t}(\phi),\ldots,\sigma_{rN,t}(\phi)\right)$ and $R_t(\phi)$ is the correlation matrix. Note that $S_t(\phi)$ and $R_t(\phi)$ are continuous in ϕ, and since eigenvalues of these matrices, $\{\sigma_{ri,t}(\phi)\}_{i=1}^N$ and $\{\lambda_{it,R}(\phi)\}_{i=1}^N$, respectively, are continuous in their elements due to the Hoffman-Wielandt Theorem, and thus are continuous in ϕ as well. Since Φ is compact by Assumption A1, S_t and R_t are positive definite, there exists $0<\underline{\sigma_r},\underline{\lambda_R}<\infty$, s.t., for all t, $\underline{\sigma_r}\leq\inf_{\phi\in\Phi}\lambda_{it,R}(\phi)$ and $\underline{\lambda_R}\leq\inf_{\phi\in\Phi}\lambda_{it,R}(\phi)$ and thus there exists $\underline{\sigma_r}^{-1}\underline{\lambda_R}^{-1}<M_h<\infty$, s.t., $\sup_{\phi\in\Phi}\|H_t^{-1}(\phi)\|\leq\underline{\sigma_r}^{-1}\underline{\lambda_R}^{-1}<M_h$. The proof of $\sup_{\phi\in\Phi}\|\tilde{H}_t^{-1}(\phi)\|\leq M_{\tilde{h}}$ follows the same argument. To prove $\sup_{\phi\in\Phi}\|\tilde{H}_t(\phi)-H_t(\phi)\|\leq\varrho^t M_{h'}$, note that a recursive substitution of model (14) leads to

$$h_t=h_0(G^*)^t+\sum_{j=0}^{t-1}\left[(G^*)^j(Q^*+A^*\text{vech}\left(r_{t-j-1}r_{t-j-1}'\right)\right].$$

Thus

$$\|\tilde{H}_t - H_t\| \leq \sqrt{\mathrm{tr}\left((\tilde{H}_t - H_t)'(\tilde{H}_t - H_t)\right)} = \|D_N(\tilde{h}_t - h_t)\|$$
$$\leq N\|A^*\|\|D_N\|\|D_N^+\|\|\tilde{h}_t - h_t\| \leq N\|A^*\|\|D_N\|\|D_N^+\|\|\tilde{h}_0 - h_0\|\|(G^*)^t\|,$$

where $\|(G^*)^t\| = \|D_N^+(G\otimes G)'\underbrace{D_N D_N^+}_{=N_N}\cdots\underbrace{D_N D_N^+}_{=N_N}(G\otimes G)'D_N\| = \|D_N^+ N_N[(G\otimes G)']^t N_N D_N\|$, since $N_N(A\otimes A)N_N = N_N(A\otimes A) = (A\otimes A)N_N$. Thus, $\|(G^*)^t\| \leq \|N_N\|^2\|D_N\|\|D_N^+\|\|[(G\otimes G)']^t\|$. Since $\rho(G\otimes G) < 1$ by Assumption A7 and Φ compact by Assumption A1, let $\varrho = \sup_{\phi\in\Phi}\rho(G(\phi)\otimes G(\phi))$ and choose $M_{h'} = N\|A^*\|\|D_N\|^2\|D_N^+\|^2\|N_N\|^2\|\tilde{h}_0 - h_0\|$, which doesn't depend on t. This completes the proof. □

Lemma 5.2. *For any $\delta > 0$, there exists an open ball $B(\phi_0, \delta)$, s.t.,*

$$\inf_{\phi\in B(\phi_0,\delta)^c\cap\Phi}\mathbb{E}\left[\tilde{l}_t(\phi)\right] > \mathbb{E}\left[\tilde{l}_t(\phi_0)\right].$$

Proof. We start by showing that $\mathbb{E}\left[\tilde{l}_t(\phi_0)\right] < \infty$ at the true parameter ϕ_0 and $-\infty < \mathbb{E}\left[\tilde{l}_t(\phi)\right]$ everywhere on Φ. Note that if $\mathbb{E}\left[\tilde{l}_t(\phi)\right] = +\infty$, the inequality holds trivially. Therefore, we assume the limit objective function is finite on Φ. Since $|\tilde{H}_t^{-1}(\phi)| \leq \|\tilde{H}_t^{-1}(\phi)\|^N \leq M_{\tilde{h}}^N < \infty$ by Lemma 5.1 and due to the elementary relation $(f_1 + f_2)^- \leq f_1^-$ if $f_2 \geq 0$, we have

$$\mathbb{E}\left[\tilde{l}_t^-(\phi)\right] = \mathbb{E}[(\underbrace{r_t'\tilde{H}_t^{-1}(\phi)r_t}_{\geq 0} + \log|\tilde{H}_t(\phi)|)^-] \leq \mathbb{E}\left[\log^-|\tilde{H}_t(\phi)|\right] < \infty.$$

At the true parameter value, since $r_t = e_t = H_t^{1/2}(\phi_0)\eta_t$, we have for some $s > 0$,

$$\mathbb{E}\left[\tilde{l}_t(\phi_0)\right] = \mathbb{E}\left[r_t'\tilde{H}_t^{-1}(\phi_0)r_t\right] + \mathbb{E}\left[\log|\tilde{H}_t(\phi_0)|\right] = \mathbb{E}[\eta_t'\eta_t] + s^{-1}\mathbb{E}\left[\log|\tilde{H}_t(\phi_0)|^s\right]$$
$$\leq \mathbb{E}\|\eta_t\|^2 + s^{-1}\log\mathbb{E}|\tilde{H}_t(\phi_0)|^s$$
$$\leq \mathbb{E}\|\eta_t\|^2 + s^{-1}\log\|\tilde{H}_t(\phi_0)\|^{sN} < \infty,$$

due to Jensen's inequality and relation $|A| \leq \|A\|^N$ for any $N\times N$ square matrix A, where the upper bound

is finite by Assumption A7 and since the return sequence is square integrable. Then, [

$$\begin{aligned}\mathbb{E}\left[\tilde{l}_t(\phi)\right] - \mathbb{E}\left[\tilde{l}_t(\phi_0)\right] &= \mathbb{E}\left[\log|\tilde{H}_t(\phi)| - \log|\tilde{H}_t(\phi_0)|\right] + \mathbb{E}\left[r_t'(\tilde{H}_t^{-1}(\phi) - \tilde{H}_t^{-1}(\phi_0))r_t\right] \\ &= \mathbb{E}\left[\log|\underbrace{\tilde{H}_t(\phi)\tilde{H}_t^{-1}(\phi_0)}_{:=\tilde{D}^{-1}(\phi)}|\right] + \mathbb{E}\left[\operatorname{tr}\left(\underbrace{\tilde{H}_t(\phi_0)\tilde{H}_t^{-1}(\phi)}_{:=\tilde{D}(\phi)} - I_N\right)\right].\end{aligned}$$

Denote the eigenvalues of matrix $\tilde{D}(\phi) = \tilde{H}_t(\phi_0)\tilde{H}_t^{-1}(\phi)$ by $\{\lambda_{it,\tilde{D}}(\phi)\}_{i=1}^N$, we have

$$\begin{aligned}\mathbb{E}\left[\tilde{l}_t(\phi)\right] - \mathbb{E}\left[\tilde{l}_t(\phi_0)\right] &= \mathbb{E}\left[-\sum_{i=1}^N \log\lambda_{it,\tilde{D}}(\phi)\right] + \mathbb{E}\left[\sum_{i=1}^N \lambda_{it,\tilde{D}}(\phi) - N\right] \\ &= \mathbb{E}\left[\sum_{i=1}^N \left(\lambda_{it,\tilde{D}}(\phi) - 1 - \log\lambda_{it,\tilde{D}}(\phi)\right)\right] \geq 0,\end{aligned}$$

where the equality holds if and only if $\lambda_{it,\tilde{D}}(\phi) - 1 = 0$ for all i and all t, a.s., which would imply $\tilde{D}(\phi) = I_N$ and $\tilde{H}_t(\phi_0) = \tilde{H}_t(\phi)$, $\forall t$ and thus $\phi = \phi_0$ by Assumption A4. Therefore, the inequality is strict for all $\phi \in B(\phi_0, \delta)^c \cap \Phi$ with strictly positive radius δ. □

Lemma 5.3. *For some $s > 1$, there exists an open neighborhood $B(\phi_0, \delta) \subset \Phi$, such that*

$$\mathbb{E}\left[\sup_{\phi\in B(\phi_0,\delta)} \|\tilde{H}_t^{-1/2}(\phi)\tilde{H}_t^{1/2}(\phi_0)\|^{2s}\right] < \infty,$$

and for any $i, j \in \{1, \ldots, d_\phi\}$,

$$\mathbb{E}\left[\sup_{\phi\in B(\phi_0,\delta)} \left\|\tilde{H}_t^{-1/2}(\phi)\frac{\partial\tilde{H}_t(\phi)}{\partial\phi_i}\tilde{H}_t^{-1/2'}(\phi)\right\|^{2s}\right] < \infty, \quad \mathbb{E}\left[\sup_{\phi\in B(\phi_0,\delta)} \left\|\tilde{H}_t^{-1/2}(\phi)\frac{\partial^2\tilde{H}_t(\phi)}{\partial\phi_i\partial\phi_j}\tilde{H}_t^{-1/2'}(\phi)\right\|^{2s}\right] < \infty.$$

Proof. First, by Assumption A7 and A8,

$$\begin{aligned}\mathbb{E}\left[\sup_{\phi\in B(\phi_0,\delta)} \|\tilde{H}_t^{-1/2}(\phi)\tilde{H}_t^{1/2}(\phi_0)\|^{2s}\right] &\leq \mathbb{E}\left[\sup_{\phi\in B(\phi_0,\delta)} \rho(\tilde{H}_t^{-1}(\phi))^s\rho(\tilde{H}_t(\phi_0))^s\right] \\ &\leq \mathbb{E}\left[\sup_{\phi\in B(\phi_0,\delta)} \operatorname{tr}\left(\tilde{H}_t^{-1}(\phi)\right)^s \operatorname{tr}\left(\tilde{H}_t(\phi_0)\right)^s\right] \\ &\leq \mathbb{E}\left[\sup_{\phi\in B(\phi_0,\delta)} N^{2s}\|\tilde{H}_t^{-1}(\phi)\|^s\|\tilde{H}_t(\phi_0)\|^s\right] \leq N^{2s}M_{\tilde{h}}^s\mathbb{E}\|\tilde{H}_t(\phi_0)\|^s < \infty,\end{aligned}$$

with constant $M_{\tilde{h}}^s$ defined analogously as in Lemma 5.1 and $M_{\tilde{h}}^s < \infty$ due to the compactness of the parameter

space. Moreover, under Assumption A8, there exists $0 < M_{\dot{h}}, M_{\ddot{h}} < \infty$, s.t., for any $i, j \in \{1, \ldots, d_\phi\}$,

$$\mathbb{E}\left[\sup_{\phi \in B(\phi_0,\delta)} \left\| \frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i} \right\|^3\right] < M_{\dot{h}} \quad \text{and} \quad \mathbb{E}\left[\sup_{\phi \in B(\phi_0,\delta)} \left\| \frac{\partial^2 \tilde{H}_t(\phi)}{\partial \phi_i \partial \phi_j} \right\|^3\right] < M_{\ddot{h}},$$

by Lemma 3 in Hafner and Preminger (2009). Thus, by Lemma 5.1 and Jensen's inequality, for any $s \in (1, \frac{3}{2}]$,

$$\begin{aligned}\mathbb{E}\left[\sup_{\phi \in B(\phi_0,\delta)} \left\| \tilde{H}_t^{-1/2}(\phi) \frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i} \tilde{H}_t^{-1/2'}(\phi) \right\|^{2s}\right] &\leq \mathbb{E}\left[\sup_{\phi \in B(\phi_0,\delta)} 2N^s \|\tilde{H}_t^{-1}(\phi)\|^s \left\| \frac{\partial \tilde{H}_t(\phi)}{\partial \phi_i} \right\|^{2s}\right] \\ &\leq 2N^s M_{\tilde{h}}^s M_{\dot{h}}^{2s/3} < \infty,\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}\left[\sup_{\phi \in B(\phi_0,\delta)} \left\| \tilde{H}_t^{-1/2}(\phi) \frac{\partial^2 \tilde{H}_t(\phi)}{\partial \phi_i \partial \phi_j} \tilde{H}_t^{-1/2'}(\phi) \right\|^{2s}\right] &\leq \mathbb{E}\left[\sup_{\phi \in B(\phi_0,\delta)} 2N^s \|\tilde{H}_t^{-1}(\phi)\|^s \left\| \frac{\partial^2 \tilde{H}_t(\phi)}{\partial \phi_i \partial \phi_j} \right\|^{2s}\right] \\ &\leq 2N^s M_{\tilde{h}}^s M_{\ddot{h}}^{2s/3} < \infty,\end{aligned}$$

which completes the proof. □